The Kalman Filter

ELG 5218 - Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Outline

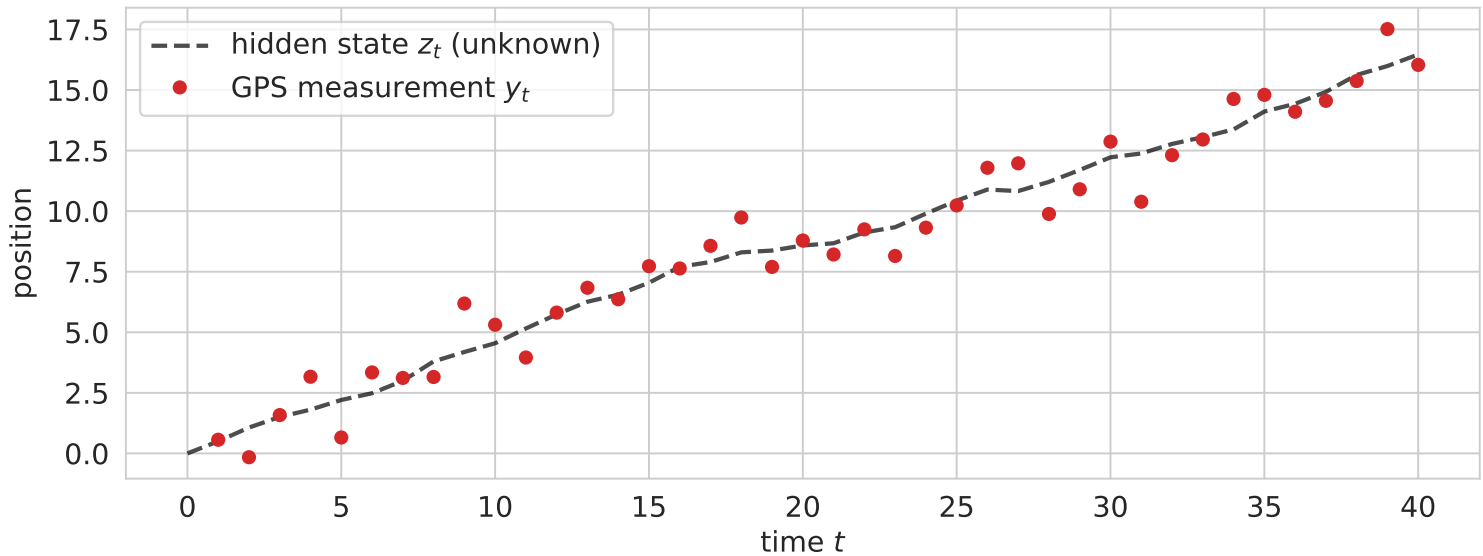
1. Introduction and Motivation
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Notebooks: `kalman_filter_interactive.ipynb` (animations used in these slides), `ssm_examples.ipynb`
Derivations are collected in the **Appendix**.

The Filtering Problem

Scenario: A robot moving in 1D space

- **State:** position z_t (hidden, uncertain)
- **Observations:** noisy GPS measurements y_t
- **Goal:** Estimate current position from all measurements so far

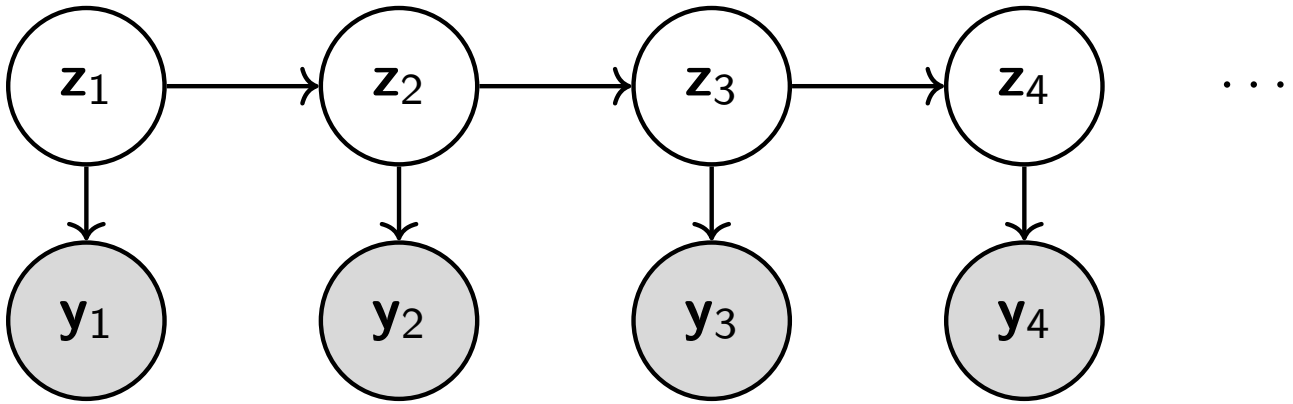


Challenge: How to optimally fuse noisy measurements with predictions from a motion model?

What is Filtering?

General Problem:

- Hidden state sequence: $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_t$
- Observations: $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_t$
- Compute posterior: $p(\mathbf{z}_t \mid \mathbf{y}_{1:t})$



Types of Inference:

- **Filtering:** $p(\mathbf{z}_t \mid \mathbf{y}_{1:t})$ — estimate current state
- **Smoothing:** $p(\mathbf{z}_t \mid \mathbf{y}_{1:T})$ — estimate past state (more data)
- **Prediction:** $p(\mathbf{z}_{t+k} \mid \mathbf{y}_{1:t})$ — forecast future

Why Kalman Filter?

The Special Case

Linear dynamics + Gaussian noise $\Rightarrow$ Exact, closed-form solution!

Advantages:

1. **Optimal:** Minimum mean squared error (MMSE) estimator
2. **Recursive:** Online processing, constant memory $O(n_z^2)$
3. **Efficient:** No integrals, just matrix operations
4. **Interpretable:** Clear predict-update structure

Historical Impact:

- Apollo lunar missions (1960s)
- GPS navigation
- Autonomous vehicles
- Economics, signal processing, control systems

Probabilistic State Space Model

Transition Model (Dynamics)

$$\mathbf{z}_t = \mathbf{F}_t \mathbf{z}_{t-1} + \mathbf{B}_t \mathbf{u}_t + \mathbf{b}_t + \mathbf{q}_t, \quad \mathbf{q}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t) \tag{1}$$

Observation Model (Measurement)

$$\mathbf{y}_t = \mathbf{H}_t \mathbf{z}_t + \mathbf{D}_t \mathbf{u}_t + \mathbf{d}_t + \mathbf{r}_t, \quad \mathbf{r}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t) \tag{2}$$

Components:

- $\mathbf{z}_t \in \mathbb{R}^{n_z}$: hidden state (position, velocity, ...)
- $\mathbf{y}_t \in \mathbb{R}^{n_y}$: observation (sensor measurement)
- $\mathbf{u}_t \in \mathbb{R}^{n_u}$: control input (known)
- $\mathbf{F}_t$: state transition matrix, $\mathbf{H}_t$: observation matrix
- $\mathbf{Q}_t$: process noise covariance, $\mathbf{R}_t$: measurement noise covariance

Two Assumptions Behind Every Filter

Joint distribution of a state-space model

$$p(\mathbf{z}_{0:t}, \mathbf{y}_{1:t}) = p(\mathbf{z}_0) \prod_{k=1}^{t} p(\mathbf{z}_k \mid \mathbf{z}_{k-1})\, p(\mathbf{y}_k \mid \mathbf{z}_k)$$

- **Markov property:** $p(\mathbf{z}_k \mid \mathbf{z}_{0:k-1}) = p(\mathbf{z}_k \mid \mathbf{z}_{k-1})$
- **Conditional independence:** $p(\mathbf{y}_k \mid \mathbf{z}_{0:k}, \mathbf{y}_{1:k-1}) = p(\mathbf{y}_k \mid \mathbf{z}_k)$

Consequence

The posterior $p(\mathbf{z}_{t-1} \mid \mathbf{y}_{1:t-1})$ summarizes *everything* the past measurements tell us about the future. $\Rightarrow$ We never need to store old data: filtering can be done **recursively**.

Bayesian Filter: Recursive Structure

Goal: Compute $p(\mathbf{z}_t \mid \mathbf{y}_{1:t})$ recursively

Initialization

Start with prior: $p(\mathbf{z}_0) = \mathcal{N}(\mathbf{z}_0 \mid \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$

Prediction Step (Chapman-Kolmogorov)

Marginalize over previous state:

$$p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1}) = \int p(\mathbf{z}_t \mid \mathbf{z}_{t-1}) p(\mathbf{z}_{t-1} \mid \mathbf{y}_{1:t-1}) d\mathbf{z}_{t-1} \tag{3}$$

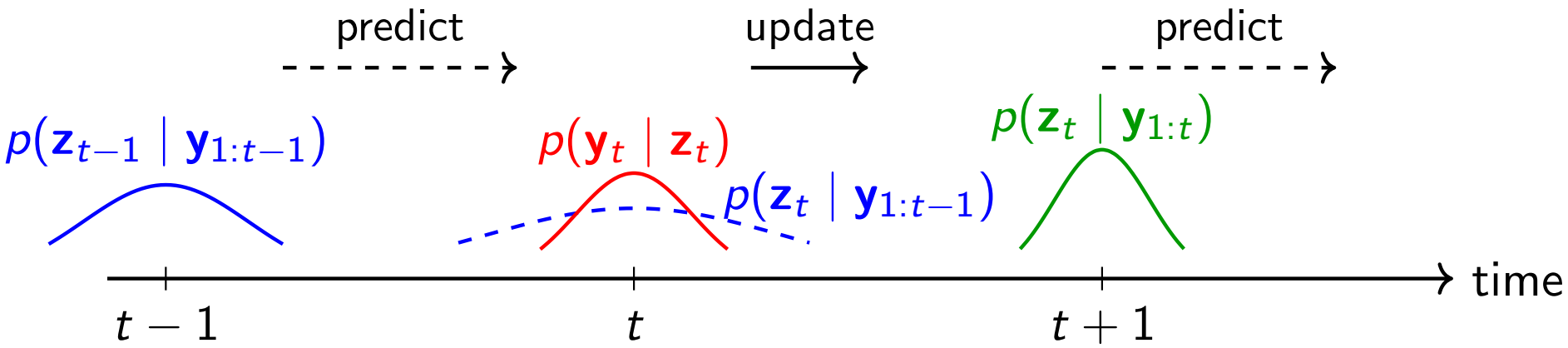
Update Step (Bayes' Rule)

Incorporate new measurement:

$$p(\mathbf{z}_t \mid \mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_t \mid \mathbf{z}_t) p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1})}{p(\mathbf{y}_t \mid \mathbf{y}_{1:t-1})} \tag{4}$$

Derivation of both steps: **Appendix A**.

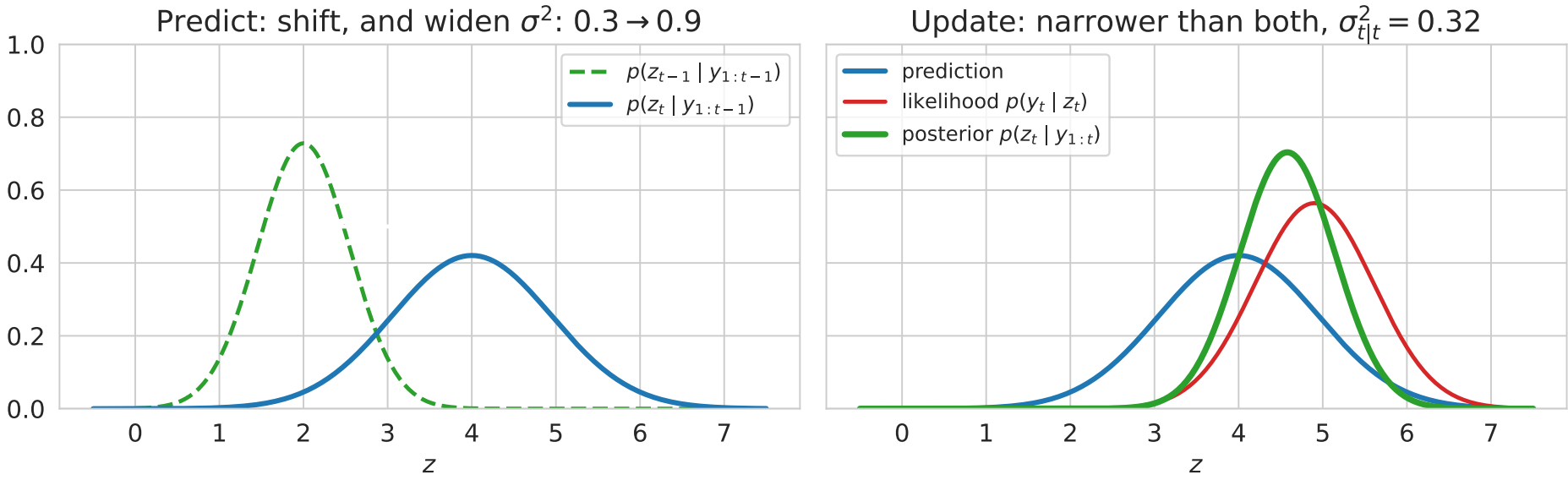
Bayesian Filter: Graphical View



Key Insight:

- **Prediction** propagates uncertainty (wider distribution)
- **Update** incorporates evidence (narrower distribution)

One Predict–Update Cycle (Notebook, Section 2)



- **Predict:** the posterior moves with the dynamics and becomes wider (process noise is added).
- **Update:** prediction × likelihood. The posterior lies **between** the two peaks, closer to the narrower one, and is **narrower than both**.
- Repeating the cycle: the widening and the narrowing balance and the width settles (steady state).

The Catch: Two Integrals

- The prediction needs an integral over $\mathbf{z}_{t-1}$; the update needs the normalizer

$$p(\mathbf{y}_t \mid \mathbf{y}_{1:t-1}) = \int p(\mathbf{y}_t \mid \mathbf{z}_t)\, p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1})\, d\mathbf{z}_t.$$

- For general nonlinear / non-Gaussian models these integrals have **no closed form** $\Rightarrow$ approximations (EKF, UKF, particle filters).

Linear dynamics + linear observations + Gaussian noise

Every distribution in the recursion stays **Gaussian**. A Gaussian is described by its mean and covariance, so the whole Bayesian filter reduces to a few matrix equations for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$:

$$\textbf{Bayesian filter} \xrightarrow{\text{linear-Gaussian}} \textbf{Kalman filter}$$

Kalman Filter: The Gaussian Case

Key Property: Linear transformations of Gaussians are Gaussian!

If $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b} + \mathbf{w}$ where $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{W})$, then:

$$\mathbf{y} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T + \mathbf{W}) \tag{5}$$

Consequence for filtering:

- If $p(\mathbf{z}_{t-1} \mid \mathbf{y}_{1:t-1}) = \mathcal{N}(\boldsymbol{\mu}_{t-1|t-1}, \boldsymbol{\Sigma}_{t-1|t-1})$
- Then $p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1})$ is also Gaussian!
- And $p(\mathbf{z}_t \mid \mathbf{y}_{1:t})$ is also Gaussian!

Kalman Filter = Recursive Gaussian Propagation

We only need to track means $\boldsymbol{\mu}_{t|t}$ and covariances $\boldsymbol{\Sigma}_{t|t}$ at each step!

Notation: $t \mid t$ vs. $t \mid t-1$

How to read the notation

"$t \mid s$" $\quad\Longrightarrow\quad$ time t given data up to time s.

State estimates (mean & covariance)

$$\boldsymbol{\mu}_{t|t-1} = \mathbb{E}[\mathbf{z}_t \mid \mathbf{y}_{1:t-1}], \qquad \boldsymbol{\Sigma}_{t|t-1} = \mathrm{Cov}(\mathbf{z}_t \mid \mathbf{y}_{1:t-1}) \quad \text{(prediction)}$$

$$\boldsymbol{\mu}_{t|t} = \mathbb{E}[\mathbf{z}_t \mid \mathbf{y}_{1:t}], \qquad \boldsymbol{\Sigma}_{t|t} = \mathrm{Cov}(\mathbf{z}_t \mid \mathbf{y}_{1:t}) \quad \text{(filtered posterior)}$$

Predicted measurement and innovation

$$\hat{\mathbf{y}}_t = \mathbb{E}[\mathbf{y}_t \mid \mathbf{y}_{1:t-1}] = \mathbf{H}_t\,\boldsymbol{\mu}_{t|t-1} + \mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t, \qquad \tilde{\mathbf{y}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t \quad \text{(innovation)}$$

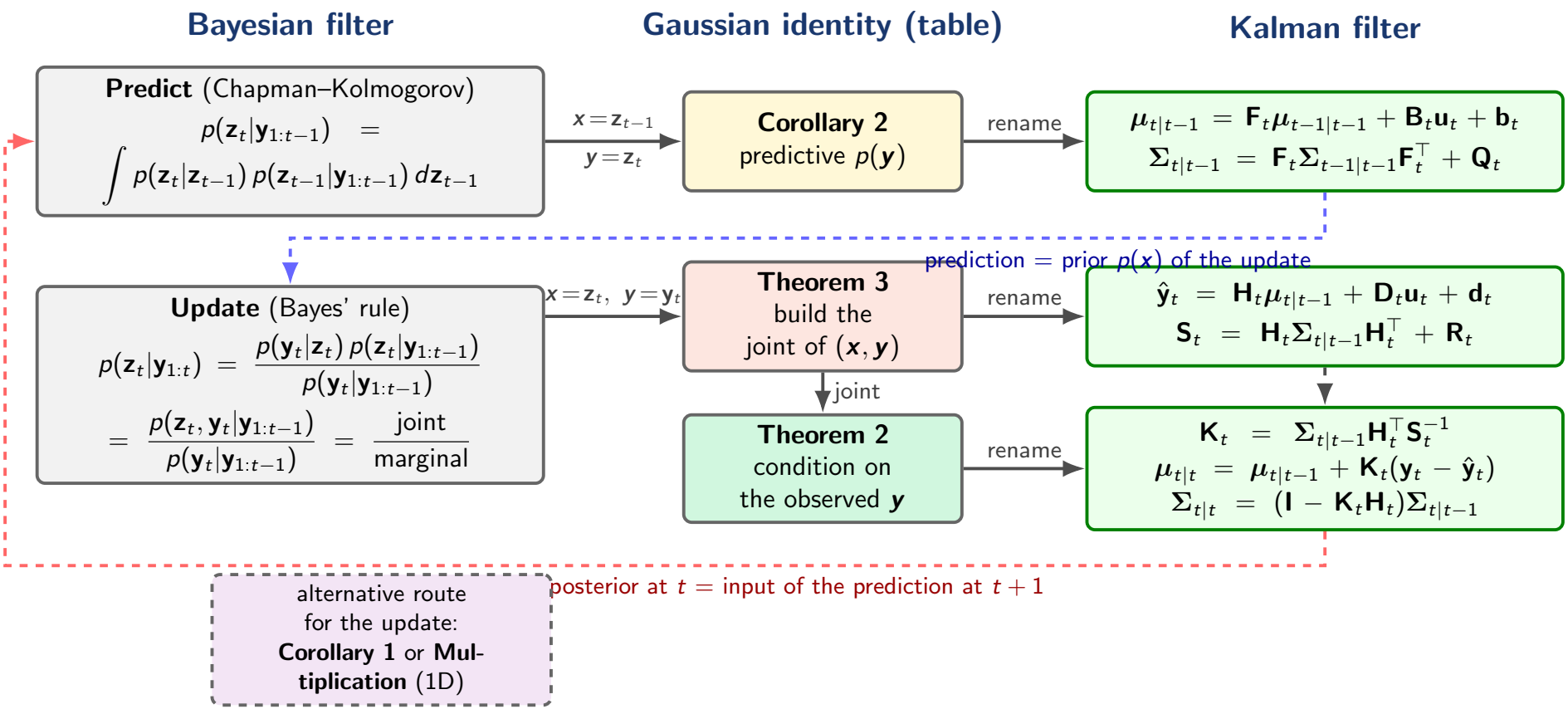
(Nonlinear: $\hat{\mathbf{y}}_t = \mathbb{E}[h(\mathbf{z}_t) \mid \mathbf{y}_{1:t-1}] \approx h(\boldsymbol{\mu}_{t|t-1})$.)

Toolbox: Gaussian Identities (Quick Reference)

Result	Setup / Conditions	Key Formulas
(A) Partitioned Gaussian: $\binom{x}{y} \sim \mathcal{N}\left(\binom{\mu_x}{\mu_y}, \binom{\Sigma_{xx}\ \Sigma_{xy}}{\Sigma_{yx}\ \Sigma_{yy}}\right)$ **(B) Affine model:** $p(x) = \mathcal{N}(x; \mu_x, \Sigma_x)$, $p(y\|x) = \mathcal{N}(y; Ax + b, \Sigma_{y\|x})$		
Theorem 1 (Marginal)	Joint Gaussian (A)	$p(x) = \mathcal{N}(x;\ \mu_x,\ \Sigma_{xx})$
Theorem 2 (Conditional)	Joint Gaussian (A)	$\mu_{x\|y} = \mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y)$, $\Sigma_{x\|y} = \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}$
Theorem 3 (Affine joint)	Affine model (B)	$p(x, y) = \mathcal{N}\left(\begin{pmatrix}\mu_x \\ A\mu_x + b\end{pmatrix}, \begin{pmatrix}\Sigma_x & \Sigma_x A^\top \\ A\Sigma_x & \Sigma_{y\|x} + A\Sigma_x A^\top\end{pmatrix}\right)$
Corollary 1 (Posterior)	Thm 2 + Thm 3	$\Sigma_{x\|y} = (\Sigma_x^{-1} + A^\top\Sigma_{y\|x}^{-1}A)^{-1}$, $\mu_{x\|y} = \Sigma_{x\|y}(\Sigma_x^{-1}\mu_x + A^\top\Sigma_{y\|x}^{-1}(y - b))$
Corollary 2 (Predictive)	Thm 1 + Thm 3	$p(y) = \mathcal{N}(y;\ A\mu_x + b,\ \Sigma_{y\|x} + A\Sigma_x A^\top)$
Multiplication (Product)	$\mathcal{N}(x; m_1, \sigma_1^2)$ times $\mathcal{N}(x; m_2, \sigma_2^2)$	$\propto \mathcal{N}(x; m, \sigma^2)$, $\sigma^2 = (\sigma_1^{-2} + \sigma_2^{-2})^{-1}$, $m = \sigma^2(m_1/\sigma_1^2 + m_2/\sigma_2^2)$

Every Kalman filter equation is one entry of this table with the symbols renamed. The next slides show the renaming.

Roadmap: Bayesian Filter → Table → Kalman Filter



Bayes' rule = joint / marginal, so the update is exactly "condition the joint Gaussian" (Theorem 2).

Bridge 1 — Prediction via Corollary 2

Bayesian prediction *what we need*

$$p(\mathbf{z}_t|\mathbf{y}_{1:t-1}) = \int \underbrace{p(\mathbf{z}_t|\mathbf{z}_{t-1})}_{\mathcal{N}(\mathbf{z}_t;\ \mathbf{F}_t\mathbf{z}_{t-1}+\mathbf{B}_t\mathbf{u}_t+\mathbf{b}_t,\ \mathbf{Q}_t)} \underbrace{p(\mathbf{z}_{t-1}|\mathbf{y}_{1:t-1})}_{\mathcal{N}(\mathbf{z}_{t-1};\ \boldsymbol{\mu}_{t-1|t-1},\ \boldsymbol{\Sigma}_{t-1|t-1})} d\mathbf{z}_{t-1}$$

Corollary 2 (predictive) *from the table*

$$p(\boldsymbol{y}) = \int \underbrace{p(\boldsymbol{y}|\boldsymbol{x})}_{\mathcal{N}(\boldsymbol{y};\ \boldsymbol{A}\boldsymbol{x}+\boldsymbol{b},\ \boldsymbol{\Sigma}_{y|x})} \underbrace{p(\boldsymbol{x})}_{\mathcal{N}(\boldsymbol{x};\ \boldsymbol{\mu}_x,\ \boldsymbol{\Sigma}_x)} d\boldsymbol{x}$$

$$\Rightarrow\ p(\boldsymbol{y}) = \mathcal{N}(\boldsymbol{y};\quad \boldsymbol{A}\boldsymbol{\mu}_x + \boldsymbol{b}\,,\qquad \boldsymbol{\Sigma}_{y|x} + \boldsymbol{A}\boldsymbol{\Sigma}_x\boldsymbol{A}^\top\ )$$

Kalman prediction *rename the coloured symbols*

$$\boldsymbol{\mu}_{t|t-1} = \mathbf{F}_t\boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t\mathbf{u}_t + \mathbf{b}_t \qquad \boldsymbol{\Sigma}_{t|t-1} = \mathbf{Q}_t + \mathbf{F}_t\boldsymbol{\Sigma}_{t-1|t-1}\mathbf{F}_t{}^\top$$

Renaming: $\boldsymbol{x} \equiv \mathbf{z}_{t-1}$, $\boldsymbol{y} \equiv \mathbf{z}_t$, $\boldsymbol{A} \equiv \mathbf{F}_t$, $\boldsymbol{b} \equiv \mathbf{B}_t\mathbf{u}_t + \mathbf{b}_t$, $\boldsymbol{\Sigma}_{y|x} \equiv \mathbf{Q}_t$, $\boldsymbol{\mu}_x \equiv \boldsymbol{\mu}_{t-1|t-1}$, $\boldsymbol{\Sigma}_x \equiv \boldsymbol{\Sigma}_{t-1|t-1}$

Bridge 2 — Update (1/2): Build the Joint via Theorem 3

Bayesian update *what we need*

Bayes' rule: $p(\mathbf{z}_t|\mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_t|\mathbf{z}_t)\,p(\mathbf{z}_t|\mathbf{y}_{1:t-1})}{p(\mathbf{y}_t|\mathbf{y}_{1:t-1})} = \frac{p(\mathbf{z}_t,\mathbf{y}_t|\mathbf{y}_{1:t-1})}{p(\mathbf{y}_t|\mathbf{y}_{1:t-1})} = \frac{\text{joint}}{\text{marginal}}$

joint built from $\underbrace{p(\mathbf{y}_t|\mathbf{z}_t)}$ $\mathcal{N}(\mathbf{y}_t;\ \mathbf{H}_t\mathbf{z}_t + \mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t,\ \mathbf{R}_t)$ and $\underbrace{p(\mathbf{z}_t|\mathbf{y}_{1:t-1})}$ $\mathcal{N}(\mathbf{z}_t;\ \boldsymbol{\mu}_{t|t-1},\ \boldsymbol{\Sigma}_{t|t-1})$

Theorem 3 (affine joint) *from the table*

given $\underbrace{p(\boldsymbol{y}|\boldsymbol{x})}$ $\mathcal{N}(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b},\ \boldsymbol{\Sigma}_{y|x})$ and $\underbrace{p(\boldsymbol{x})}$ $\mathcal{N}(\boldsymbol{\mu}_x,\ \boldsymbol{\Sigma}_x)$

$$\Rightarrow \begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{y}\end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix}\boldsymbol{\mu}_x\\ \boldsymbol{A}\boldsymbol{\mu}_x + \boldsymbol{b}\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma}_x & \boldsymbol{\Sigma}_x\boldsymbol{A}^\top\\ \boldsymbol{A}\boldsymbol{\Sigma}_x & \boldsymbol{\Sigma}_{y|x} + \boldsymbol{A}\boldsymbol{\Sigma}_x\boldsymbol{A}^\top\end{pmatrix}\right)$$

Kalman: joint of $\mathbf{z}_t$ and $\mathbf{y}_t$ *rename, and name the blocks*

$\hat{\mathbf{y}}_t = \mathbf{H}_t\boldsymbol{\mu}_{t|t-1} + \mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t$ and $\mathbf{S}_t = \mathbf{R}_t + \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t{}^\top$

Renaming: $\boldsymbol{x} \equiv \mathbf{z}_t$, $\boldsymbol{y} \equiv \mathbf{y}_t$, $\boldsymbol{A} \equiv \mathbf{H}_t$, $\boldsymbol{b} \equiv \mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t$, $\boldsymbol{\Sigma}_{y|x} \equiv \mathbf{R}_t$, $\boldsymbol{\mu}_x \equiv \boldsymbol{\mu}_{t|t-1}$, $\boldsymbol{\Sigma}_x \equiv \boldsymbol{\Sigma}_{t|t-1}$; off-diagonal block: $\mathrm{Cov}(\mathbf{z}_t, \mathbf{y}_t) = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top$.

Bridge 3 — Update (2/2): Condition via Theorem 2

Joint from Bridge 2 *input; then $\mathbf{y}_t$ is observed*

$$\begin{pmatrix} \mathbf{z}_t \\ \mathbf{y}_t \end{pmatrix} \Big| \, \mathbf{y}_{1:t-1} \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_{t|t-1} \\ \hat{\mathbf{y}}_t \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{t|t-1} & \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \\ \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1} & \mathbf{S}_t \end{pmatrix} \right)$$

Theorem 2 (conditional) *from the table*

$$\boldsymbol{\mu}_{x|y} = \boldsymbol{\mu}_x + \underbrace{\boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}}_{\text{regression coefficient}}(\boldsymbol{y} - \boldsymbol{\mu}_y)$$

$$\boldsymbol{\Sigma}_{x|y} = \boldsymbol{\Sigma}_{xx} - \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}$$

Kalman update *rename; call the regression coefficient the gain*

$$\mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top\mathbf{S}_t^{-1} \qquad \boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t(\mathbf{y}_t - \hat{\mathbf{y}}_t)$$

$$\boldsymbol{\Sigma}_{t|t} = \boldsymbol{\Sigma}_{t|t-1} - \mathbf{K}_t\mathbf{S}_t\mathbf{K}_t^\top = (\mathbf{I} - \mathbf{K}_t\mathbf{H}_t)\boldsymbol{\Sigma}_{t|t-1}$$

Renaming: $\boldsymbol{x} \equiv \mathbf{z}_t$, $\boldsymbol{y} \equiv \mathbf{y}_t$, $\boldsymbol{\mu}_x \equiv \boldsymbol{\mu}_{t|t-1}$, $\boldsymbol{\mu}_y \equiv \hat{\mathbf{y}}_t$, $\boldsymbol{\Sigma}_{xx} \equiv \boldsymbol{\Sigma}_{t|t-1}$, $\boldsymbol{\Sigma}_{xy} \equiv \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top$, $\boldsymbol{\Sigma}_{yy} \equiv \mathbf{S}_t$.

Last step: $\boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx} = \mathbf{K}_t\mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1} = \mathbf{K}_t\mathbf{S}_t\mathbf{K}_t^\top$.

Prediction Step: What the Equations Mean

Apply the Gaussian property to the dynamics $\mathbf{z}_t = \mathbf{F}_t\mathbf{z}_{t-1} + \mathbf{B}_t\mathbf{u}_t + \mathbf{b}_t + \mathbf{q}_t$:

Prediction (time update): $p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1}) = \mathcal{N}(\boldsymbol{\mu}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1})$

$$\boldsymbol{\mu}_{t|t-1} = \underbrace{\mathbf{F}_t\boldsymbol{\mu}_{t-1|t-1}}_{\text{move with the model}} + \underbrace{\mathbf{B}_t\mathbf{u}_t + \mathbf{b}_t}_{\text{known inputs}}$$

$$\boldsymbol{\Sigma}_{t|t-1} = \underbrace{\mathbf{F}_t\boldsymbol{\Sigma}_{t-1|t-1}\mathbf{F}_t^\top}_{\text{propagated uncertainty}} + \underbrace{\mathbf{Q}_t}_{\text{new uncertainty}}$$

Intuition

- The mean simply follows the motion model; known inputs add no uncertainty.
- $\mathbf{F}_t$ *reshapes* the uncertainty (stretches, rotates, correlates); $\mathbf{Q}_t$ *inflates* it.
- Without measurements, prediction alone makes $\boldsymbol{\Sigma}$ grow step after step.

Where it comes from: **Bridge 1** (Corollary 2).

Update Step: What the Equations Mean

Condition the Gaussian prediction on the new measurement $\mathbf{y}_t = \mathbf{H}_t\mathbf{z}_t + \mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t + \mathbf{r}_t$:

Update (measurement update)

$$\hat{\mathbf{y}}_t = \mathbf{H}_t\boldsymbol{\mu}_{t|t-1} + \mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t \qquad \text{predicted measurement}$$

$$\mathbf{S}_t = \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top + \mathbf{R}_t \qquad \text{its uncertainty}$$

$$\mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top\mathbf{S}_t^{-1} \qquad \text{Kalman gain}$$

$$\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t\left(\mathbf{y}_t - \hat{\mathbf{y}}_t\right) \qquad \text{correct by the innovation}$$

$$\boldsymbol{\Sigma}_{t|t} = \left(\mathbf{I} - \mathbf{K}_t\mathbf{H}_t\right)\boldsymbol{\Sigma}_{t|t-1} \qquad \text{uncertainty shrinks}$$

Intuition

- Compare what we *expected* to measure ($\hat{\mathbf{y}}_t$) with what we *did* measure ($\mathbf{y}_t$).
- Move the estimate by a fraction $\mathbf{K}_t$ of the surprise $\tilde{\mathbf{y}}_t$.
- $\mathbf{K}_t$ is large when the prediction is uncertain compared with the sensor.

Where it comes from: **Bridges 2–3** (Theorem 3, then Theorem 2).

Kalman Filter: Complete Algorithm

Initialization

$\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0$ (prior mean and covariance)

For $t = 1, 2, \ldots, T$:

Prediction Step

$$\boldsymbol{\mu}_{t|t-1} = \mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t \mathbf{u}_t + \mathbf{b}_t \tag{6}$$

$$\boldsymbol{\Sigma}_{t|t-1} = \mathbf{F}_t \boldsymbol{\Sigma}_{t-1|t-1} \mathbf{F}_t^T + \mathbf{Q}_t \tag{7}$$

Update Step

$$\hat{\mathbf{y}}_t = \mathbf{H}_t \boldsymbol{\mu}_{t|t-1} + \mathbf{D}_t \mathbf{u}_t + \mathbf{d}_t \tag{8}$$

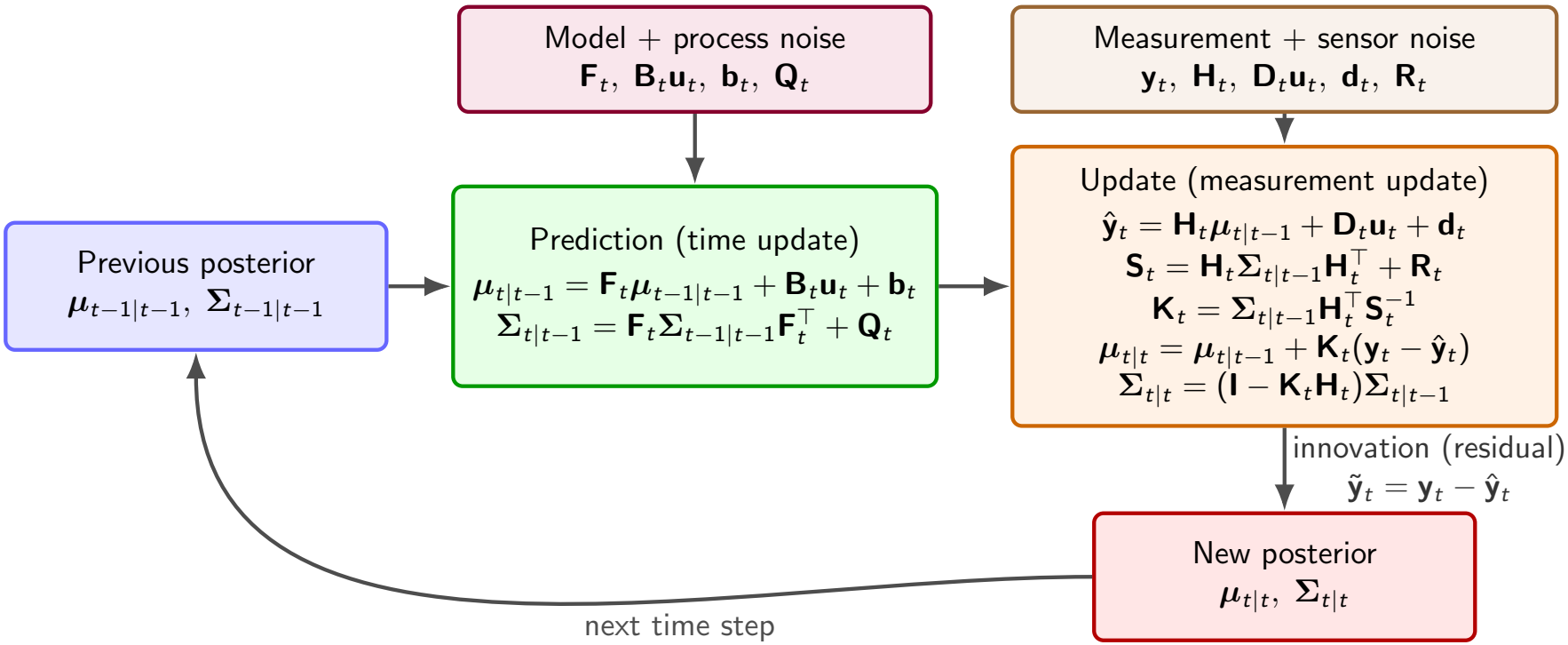
$$\mathbf{S}_t = \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^T + \mathbf{R}_t \tag{9}$$

$$\mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^T \mathbf{S}_t^{-1} \tag{10}$$

$$\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t (\mathbf{y}_t - \hat{\mathbf{y}}_t) \tag{11}$$

$$\boldsymbol{\Sigma}_{t|t} = (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t) \boldsymbol{\Sigma}_{t|t-1} \tag{12}$$

Kalman Filter Operations (One Time Step)



Kalman Filter: 1D Scalar Case

Model (q_t, r_t: noise variances)

$$z_t = z_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, q_t)$$
$$y_t = z_t + v_t, \quad v_t \sim \mathcal{N}(0, r_t)$$

Replaced from the matrix KF

- $\boldsymbol{\mu}_{t|t} \to \mu_{t|t}, \quad \boldsymbol{\Sigma}_{t|t} \to \sigma^2_{t|t}$
- $F_t = 1$ so $F_t^2 \sigma^2 \to \sigma^2$
- $H_t = 1$ so $S_t = H_t^2 \sigma^2 + r_t \to \sigma^2 + r_t$
- $u_t = b_t = d_t = 0$ so $\hat{y}_t = H_t \mu_{t|t-1} \to \mu_{t|t-1}$
- $\mathbf{K}_t = \Sigma H^\top S^{-1} \to K_t = \sigma^2_{t|t-1} / (\sigma^2_{t|t-1} + r_t)$

Intuition: q_t makes uncertainty grow over time (prediction), r_t controls how much we trust the measurement (update).

Complete algorithm

Initialization: $z_0 \sim \mathcal{N}(\mu_0, \sigma_0^2)$
For $t = 1, \ldots, T$**:**

Prediction (time update):

$$\mu_{t|t-1} = \mu_{t-1|t-1} \qquad \sigma^2_{t|t-1} = \sigma^2_{t-1|t-1} + q_t$$

Update (measurement update):

$$\hat{y}_t = \mu_{t|t-1} \qquad S_t = \sigma^2_{t|t-1} + r_t$$

$$K_t = \frac{\sigma^2_{t|t-1}}{\sigma^2_{t|t-1} + r_t}$$

$$\mu_{t|t} = \mu_{t|t-1} + K_t (y_t - \hat{y}_t)$$

$$\sigma^2_{t|t} = (1 - K_t) \sigma^2_{t|t-1}$$

1D Shortcut: The Update Is a Product of Two Gaussians

Bayesian update, 1D ($H_t = 1$) *what we need*

$$p(z_t|y_{1:t}) \propto \underbrace{p(z_t|y_{1:t-1})}_{\mathcal{N}(z_t;\ \mu_{t|t-1},\ \sigma^2_{t|t-1})} \times \underbrace{p(y_t|z_t)}_{\mathcal{N}(y_t;\ z_t,\ r_t) = \mathcal{N}(z_t;\ y_t,\ r_t)}$$

Multiplication (product) *from the table*

$$\mathcal{N}(x; m_1, \sigma_1^2) \times \mathcal{N}(x; m_2, \sigma_2^2)$$

$$\sigma^2 = \left(\sigma_1^{-2} + \sigma_2^{-2}\right)^{-1} \qquad m = \sigma^2\left(m_1/\sigma_1^2 + m_2/\sigma_2^2\right)$$

1D Kalman update *rename and simplify*

$$\sigma^2_{t|t} = \frac{\sigma^2_{t|t-1}\, r_t}{\sigma^2_{t|t-1} + r_t} = (1 - K_t)\, \sigma^2_{t|t-1}$$

$$\mu_{t|t} = \frac{r_t\, \mu_{t|t-1} + \sigma^2_{t|t-1}\, y_t}{\sigma^2_{t|t-1} + r_t} = \mu_{t|t-1} + K_t(y_t - \mu_{t|t-1})$$

with $K_t = \sigma^2_{t|t-1}/(\sigma^2_{t|t-1} + r_t)$. 1D prediction is Corollary 2 with $A = 1$, $b = 0$, $\Sigma_{y|x} = q_t$.

1D Kalman Filter: A Worked Step

Model: random walk + noisy measurement

$$z_t = z_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, 0.2^2), \qquad y_t = z_t + v_t, \quad v_t \sim \mathcal{N}(0, 0.5^2).$$

Assume at $t-1$: $\mu_{t-1|t-1} = 2.0, \quad \sigma^2_{t-1|t-1} = 0.3^2 = 0.09$

Predict

$\mu_{t|t-1} = 2.0$

$\sigma^2_{t|t-1} = 0.09 + 0.04 = 0.13$

$\hat{y}_t = \mu_{t|t-1} = 2.0$

Measurement arrives

$y_t = 2.6$, innovation $\tilde{y}_t = y_t - \hat{y}_t = 0.6$

Update

$S_t = 0.13 + 0.25 = 0.38$

$K_t = 0.13/0.38 = 0.342$

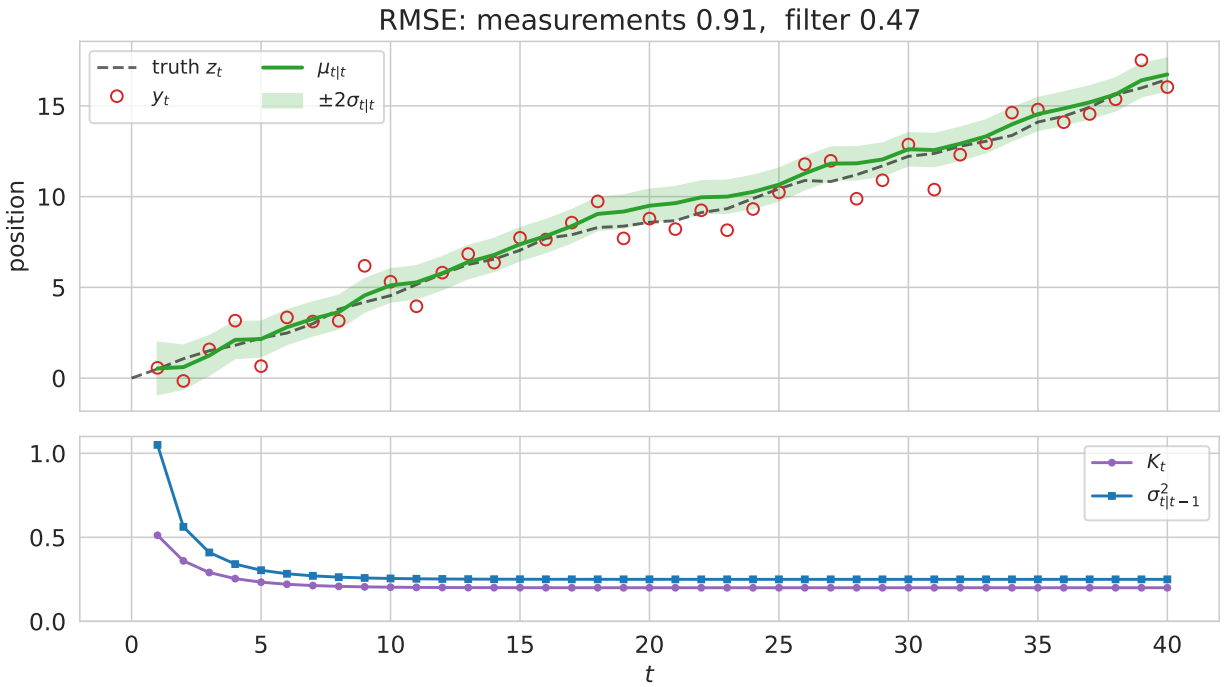
$\mu_{t|t} = 2.0 + 0.342 \cdot 0.6 = 2.205$

$\sigma^2_{t|t} = (1 - K_t)\, 0.13 = 0.0856$

Observe: $0.0856 < \min(0.13,\ 0.25)$ — the posterior is more certain than *both* the prediction and the measurement. Changing y_t would change $\mu_{t|t}$ but **not** K_t or $\sigma^2_{t|t}$.

1D Example: Tracking the Robot

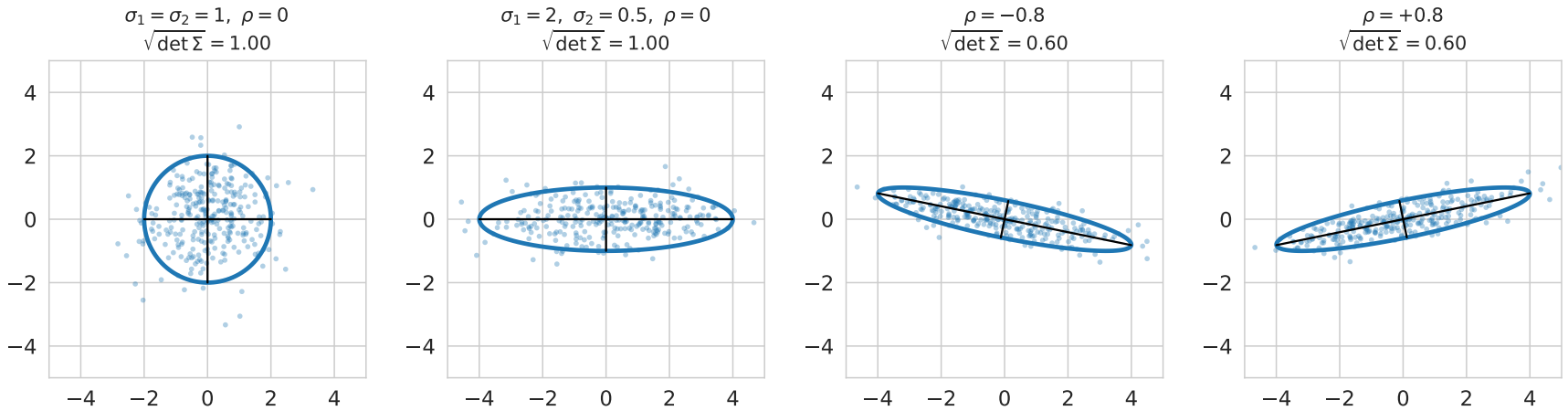
$z_t = z_{t-1} + s\,\Delta t + w_t, \;\; y_t = z_t + v_t \;\Rightarrow\; F_t = 1,\; b_t = s\Delta t,\; Q_t = q = 0.05,\; H_t = 1,\; R_t = r = 1$



- The filter roughly **halves** the error of the raw measurements.
- K_t starts at 0.5 (prior $\sigma_0^2 = 1$) and settles at ≈ 0.2 within a few steps; the $\pm 2\sigma$ band becomes constant.
- $1/K_\infty \approx 5$: the filter behaves like an average over the last $\sim$5 measurements, while following the known drift exactly.

Reading a Covariance Matrix: Uncertainty Ellipses

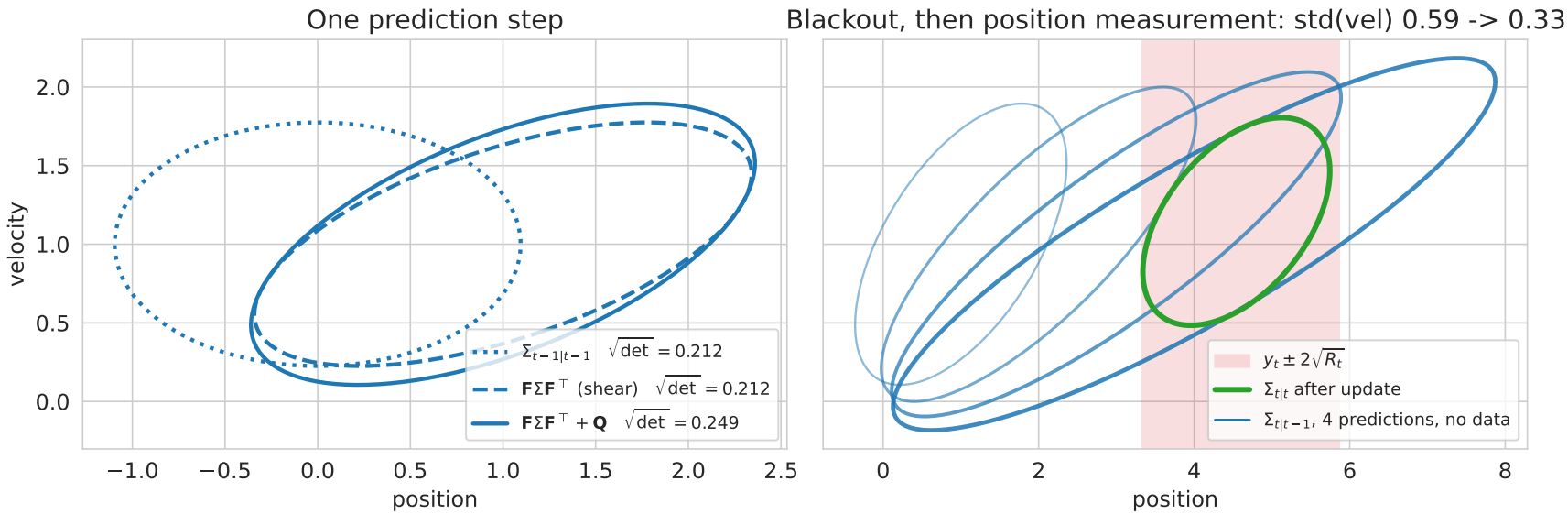
$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\,\sigma_1\sigma_2 \\ \rho\,\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$ drawn as the 2σ ellipse of $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$



- Diagonal entries stretch the ellipse along the axes; the correlation ρ **tilts** it.
- Axes = eigenvectors of $\boldsymbol{\Sigma}$, half-lengths $2\sqrt{\lambda_i}$.
- Area $\propto \sqrt{\det \boldsymbol{\Sigma}} = \sigma_1\sigma_2\sqrt{1-\rho^2}$: correlation *reduces* joint uncertainty — knowing one variable tells us about the other.

What Prediction and Update Do to Σ

Constant-velocity model, $\mathbf{z}_t = [\text{position}, \text{velocity}]^\top$, $\mathbf{F}_t = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$, only position measured: $\mathbf{H}_t = [1 \quad 0]$.



- $\mathbf{F}_t\Sigma\mathbf{F}_t^\top$ **shears**: faster states move further, so position and velocity become correlated. Since $\det \mathbf{F}_t = 1$, the area is unchanged.
- $+\mathbf{Q}_t$ **inflates**: this is the only source of new uncertainty.
- Without measurements the ellipse keeps growing; one position measurement collapses it in position **and** in velocity.

Why Measuring Position Also Reduces Velocity Uncertainty

For $\mathbf{H}_t = [1 \;\; 0]$ and $\boldsymbol{\Sigma}_{t|t-1} = \begin{bmatrix} \Sigma_{pp} & \Sigma_{pv} \\ \Sigma_{pv} & \Sigma_{vv} \end{bmatrix}$:

$$S_t = \Sigma_{pp} + R_t, \qquad \mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top S_t^{-1} = \begin{bmatrix} K_p \\ K_v \end{bmatrix} = \frac{1}{\Sigma_{pp} + R_t} \begin{bmatrix} \Sigma_{pp} \\ \Sigma_{pv} \end{bmatrix}$$

$$\Sigma_{vv,\,t|t} = \Sigma_{vv} - \frac{\Sigma_{pv}^2}{\Sigma_{pp} + R_t} = \Sigma_{vv} - K_v^2 S_t, \qquad \frac{\Sigma_{vv,\,t|t}}{\Sigma_{vv}} = 1 - \rho^2 K_p$$

with the position–velocity correlation $\rho = \Sigma_{pv} / \sqrt{\Sigma_{pp}\Sigma_{vv}}$

- The **velocity gain** K_v is *linear* in the cross-covariance Σ_{pv} (created by the prediction step): it sets the direction and size of the velocity correction $K_v \tilde{y}_t$.
- The **velocity-variance reduction** $\Sigma_{pv}^2/(\Sigma_{pp} + R_t)$ is *quadratic* in Σ_{pv}: positive and negative correlation remove the same amount; the fraction removed is $\rho^2 K_p$.
- If $\Sigma_{pv} = 0$ ($\rho = 0$), measuring position tells us nothing about velocity.

Take-away

Hidden (unmeasured) states are estimated through the correlations encoded in $\boldsymbol{\Sigma}$.

The Kalman Gain: Optimal Weighting

$$\mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^T(\mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^T + \mathbf{R}_t)^{-1}$$

Interpretation:

$$\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t \underbrace{(\mathbf{y}_t - \hat{\mathbf{y}}_t)}_{\text{innovation } \tilde{\mathbf{y}}_t}$$

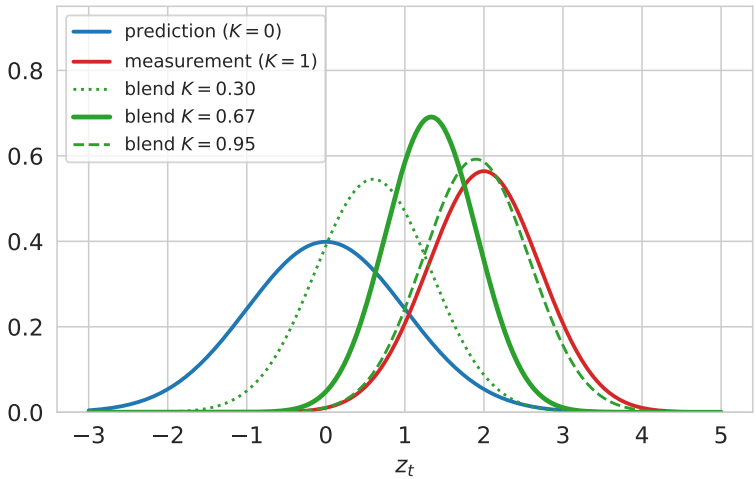
- $\mathbf{K}_t$ determines how much we trust the new measurement vs. the prediction
- Innovation $\tilde{\mathbf{y}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$: "surprise" in the measurement
- Large $\mathbf{K}_t \rightarrow$ trust measurement more
- Small $\mathbf{K}_t \rightarrow$ trust prediction more

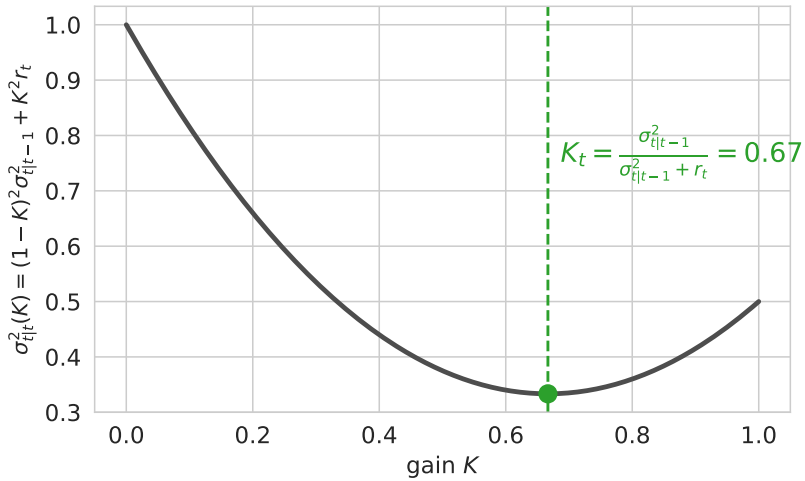
Four ways to understand it (next slides): optimal blend (MMSE) · ratio of uncertainties · regression slope · data-independent schedule.

Why This Gain? The Minimum-Variance Blend

Scalar case, $H_t = 1$: any blend $\mu_{t|t} = (1-K)\,\mu_{t|t-1} + K\,y_t$ looks reasonable. With independent errors,

$$\sigma^2_{t|t}(K) = (1-K)^2\,\sigma^2_{t|t-1} + K^2\,r_t, \qquad \frac{d\sigma^2_{t|t}}{dK} = 0 \;\Rightarrow\; K_t = \frac{\sigma^2_{t|t-1}}{\sigma^2_{t|t-1} + r_t}.$$

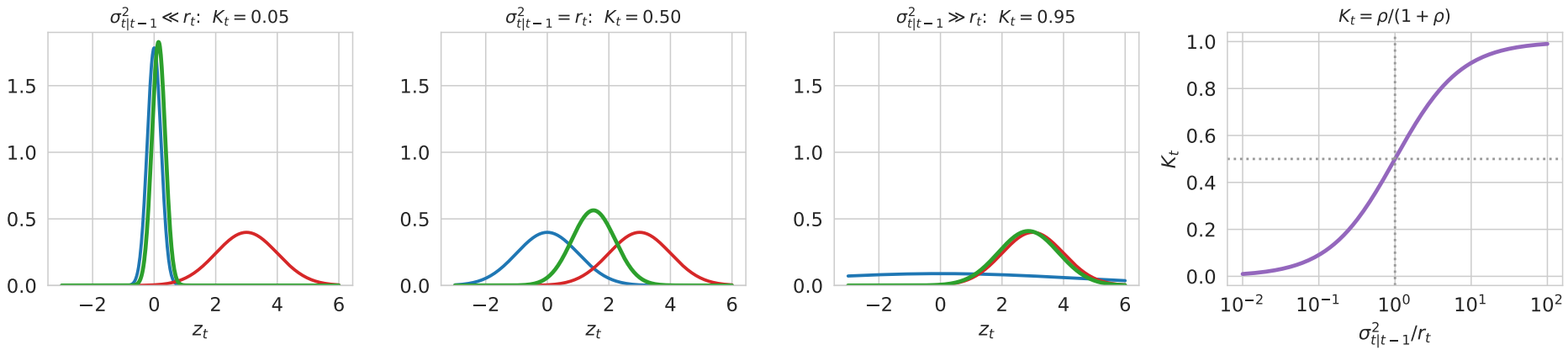




- $K = 0$: keep the prediction (error $\sigma^2_{t|t-1}$); $K = 1$: take the measurement (error r_t).
- In between, independent errors partly cancel; the minimum is the Kalman gain, with $\sigma^2_{t|t} = (1-K_t)\sigma^2_{t|t-1}$.
- Matrix version (minimize the trace): **Appendix B.4** — this is the MMSE property.

The Gain Depends Only on Relative Uncertainty

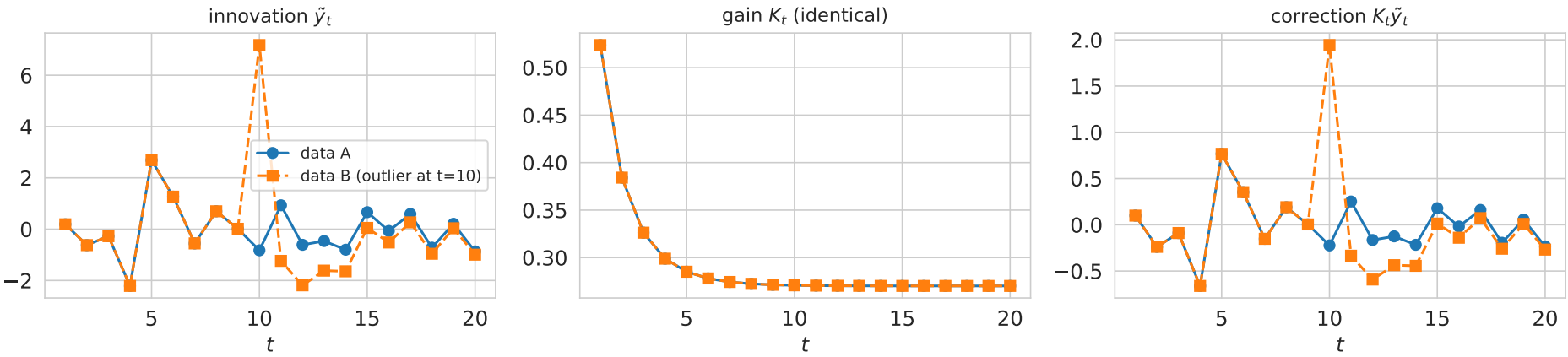
$$K_t = \frac{\sigma^2_{t|t-1}}{\sigma^2_{t|t-1} + r_t} = \frac{\rho}{1+\rho}, \qquad \rho = \frac{\sigma^2_{t|t-1}}{r_t}$$



- $\sigma^2_{t|t-1} \ll r_t$ (confident prediction, noisy sensor): $K_t \approx 0 \Rightarrow \mu_{t|t} \approx \mu_{t|t-1}$
- $\sigma^2_{t|t-1} = r_t$: $K_t = 0.5 \Rightarrow \mu_{t|t} = 0.5\,\mu_{t|t-1} + 0.5\,y_t$
- $\sigma^2_{t|t-1} \gg r_t$ (uncertain prediction, accurate sensor): $K_t \approx 1 \Rightarrow \mu_{t|t} \approx y_t$
- Doubling both variances leaves K_t unchanged; only the posterior width doubles.

A Common Misconception: "A Big Surprise Increases the Gain"

Same filter, two data sets that differ only by an outlier at $t = 10$:



- $\mathbf{K}_t$ depends on $\boldsymbol{\Sigma}_{t|t-1}$, $\mathbf{H}_t$, $\mathbf{R}_t$ only — **never on $\tilde{\mathbf{y}}_t$.** The two gain sequences are identical.
- A surprise produces a large *correction* $\mathbf{K}_t\tilde{\mathbf{y}}_t$, not a different gain.
- **The estimate reacts linearly:** $\mu_{t|t} = \mu_{t|t-1} + K_t\tilde{y}_t$ moves by $0.27 \times 8 \approx 2.2$; the error then decays by $(1 - K) \approx 0.73$ per step.
- **This reaction is the "sensitivity to outliers":** the outlier is $6\sqrt{S_t}$ from the prediction (impossible under the model) but gets the same weight as any measurement, so its influence grows with its size.

Extreme Cases

Perfect measurements: $\mathbf{R}_t \to \mathbf{0}$

$\mathbf{K}_t\mathbf{H}_t \to \mathbf{I}$ on the measured subspace; if $\mathbf{H}_t$ is square and invertible, $\mathbf{K}_t \to \mathbf{H}_t^{-1}$ and $\boldsymbol{\mu}_{t|t} \to \mathbf{H}_t^{-1}\mathbf{y}_t$: trust the measurement fully.

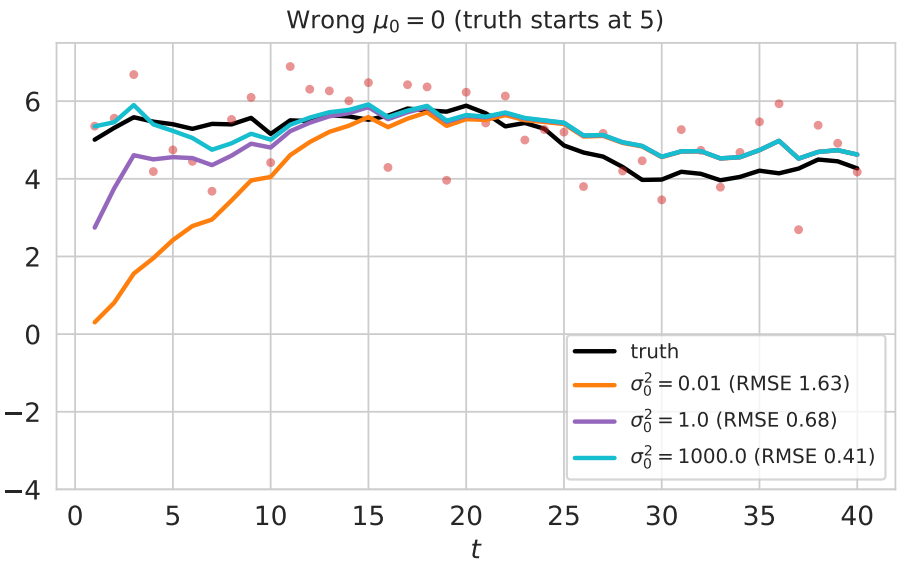
Perfect model: $\mathbf{Q}_t \to \mathbf{0}$

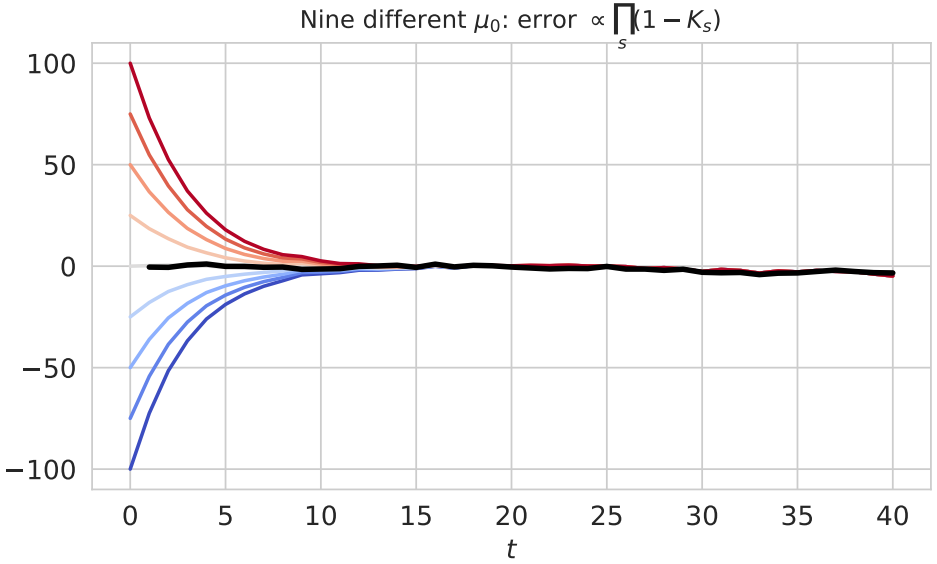
Prediction adds no uncertainty, so $\boldsymbol{\Sigma}_{t|t}$ keeps shrinking and $\mathbf{K}_t \to \mathbf{0}$ as $t \to \infty$ (scalar: $K_t = \sigma_0^2/(t\,\sigma_0^2 + r) \approx 1/t$, a running average). Early gains can still be large if $\boldsymbol{\Sigma}_0$ is large.

Practical consequence

A filter with $\mathbf{Q}_t = \mathbf{0}$ eventually stops listening to measurements. A small positive $\mathbf{Q}_t$ is often added on purpose, even for nearly static states.

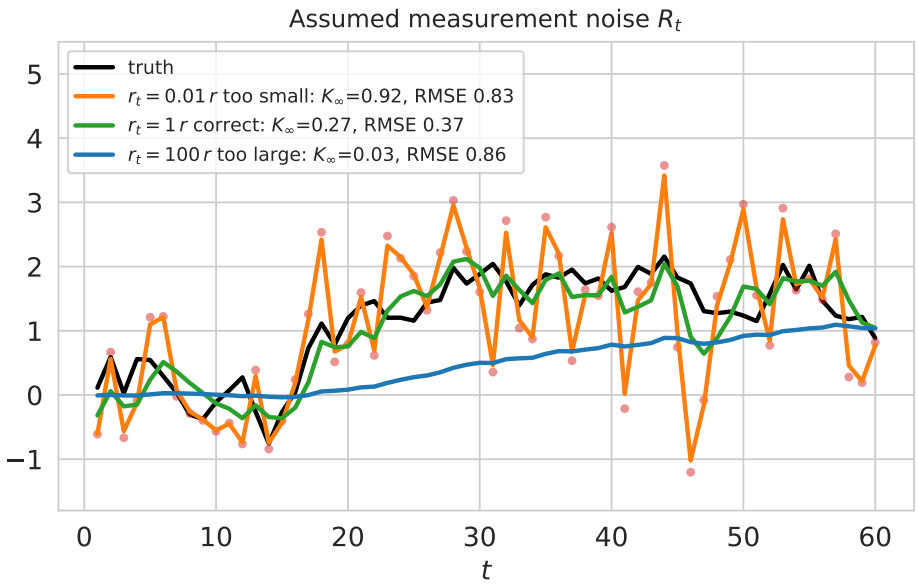
Initial Conditions: Σ_0 and μ_0

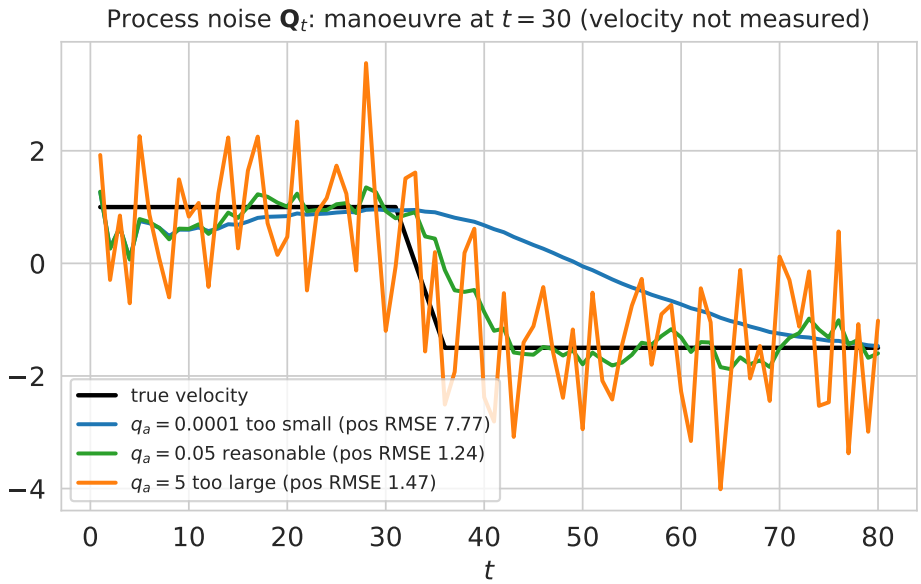




- **Left:** with a wrong μ_0, an overconfident $\sigma_0^2 = 0.01$ gives $K_1 \approx 0.06$ and needs $\sim$15 steps to unlearn the prior; a diffuse $\sigma_0^2 = 1000$ gives $K_1 \approx 1$ and jumps onto the first measurement. **When in doubt, choose Σ_0 large.**
- **Right:** filters that differ only in μ_0 obey $e_t = (1 - K_t)F_t\, e_{t-1}$, so the initial error is **forgotten geometrically** (from 100 to $\approx 10^{-4}$ in 40 steps).
- Both effects are transients — provided the model is right and $K_\infty > 0$.

Choosing $\mathbf{R}_t$ and $\mathbf{Q}_t$





- $\mathbf{R}_t$ **too small:** $K \approx 1$, the estimate copies the measurement noise. **Too large:** $K \approx 0$, smooth but lagging. The true value gives the lowest RMSE.
- $\mathbf{Q}_t$ **too small:** the filter "knows" the velocity cannot change and lags badly after a manoeuvre. **Too large:** reacts instantly but is noisy.
- $\mathbf{Q}_t$ and $\mathbf{R}_t$ act only through $\mathbf{K}_t$: together they set the filter's **bandwidth** (responsiveness vs. noise suppression).

Checking a Filter Without Ground Truth: the NIS

Is the measurement farther from the prediction than the filter's uncertainty would suggest?

$$\underbrace{\tilde{\mathbf{y}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t}_{\text{measurement prediction error}} \quad , \quad \underbrace{\mathbf{S}_t = \mathrm{Cov}(\mathbf{y}_t \mid \mathbf{y}_{1:t-1})}_{\text{predicted uncertainty}}.$$

Normalized Innovation Squared (NIS)

$$\epsilon_t = \tilde{\mathbf{y}}_t^\top \mathbf{S}_t^{-1} \tilde{\mathbf{y}}_t, \qquad \text{scalar:} \quad \epsilon_t = \left(\frac{\tilde{y}_t}{\sqrt{S_t}} \right)^2.$$

For one measurement, NIS is the **squared number of predicted standard deviations** between the measurement and its prediction.

Example: If $\tilde{y}_t = 3\,\mathrm{m}$ and $\sqrt{S_t} = 2\,\mathrm{m}$, then $\epsilon_t = (3/2)^2 = 2.25$. The error is 1.5 predicted standard deviations.

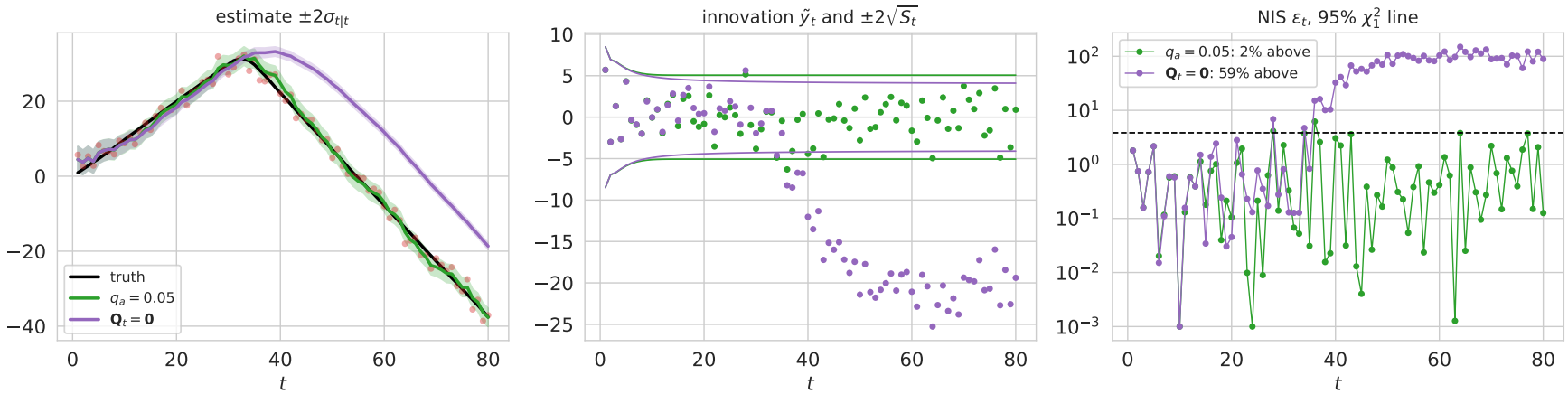
How to interpret the values

- Under a correct linear Gaussian model with consistent initialization, $\epsilon_t \sim \chi^2_{n_y}$ and $\mathbb{E}[\epsilon_t] = n_y$, where n_y is the number of measurement components.
- For $n_y = 1$, about 95% of values should satisfy $\epsilon_t \leq 3.84$, equivalently $|\tilde{y}_t| \leq 1.96\sqrt{S_t}$. Occasional exceedances are expected.
- Over many steps: $\bar{\epsilon} \gg n_y$ suggests underestimated uncertainty, bias, or model mismatch; $\bar{\epsilon} \ll n_y$ suggests overestimated uncertainty.

No true state is needed: compare the measurement with its prediction *before the update*.

Innovations, Consistency and Divergence

Manoeuvring target; filter with $q_a = 0.05$ vs. filter assuming a perfect model ($\mathbf{Q}_t = \mathbf{0}$):



- With $\mathbf{Q}_t = \mathbf{0}$: $\mathbf{\Sigma}_{t|t} \to \mathbf{0}$, $\mathbf{K}_t \to \mathbf{0}$; after the manoeuvre the filter is **confidently wrong** (thin band, far from the truth).
- Under a correct model $\tilde{\mathbf{y}}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_t)$, white, and NIS $\epsilon_t = \tilde{\mathbf{y}}_t^\top \mathbf{S}_t^{-1} \tilde{\mathbf{y}}_t \sim \chi^2_{n_y}$: about 5% above the 95% line (here 2% vs. 59%).
- The truth is unknown in practice: **innovation monitoring is the real-time health check**.

Example 2: Constant Velocity Model (2D Tracking)

State: $\mathbf{z}_t = [x_t, \dot{x}_t, y_t, \dot{y}_t]^T$ (position + velocity in 2D)
Dynamics: Newton's laws with random acceleration

$$\mathbf{F}_t = \begin{bmatrix} 1 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{Q}_t = q \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} & 0 & 0 \\ \frac{\Delta t^3}{2} & \Delta t^2 & 0 & 0 \\ 0 & 0 & \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} \\ 0 & 0 & \frac{\Delta t^3}{2} & \Delta t^2 \end{bmatrix} \tag{13}$$

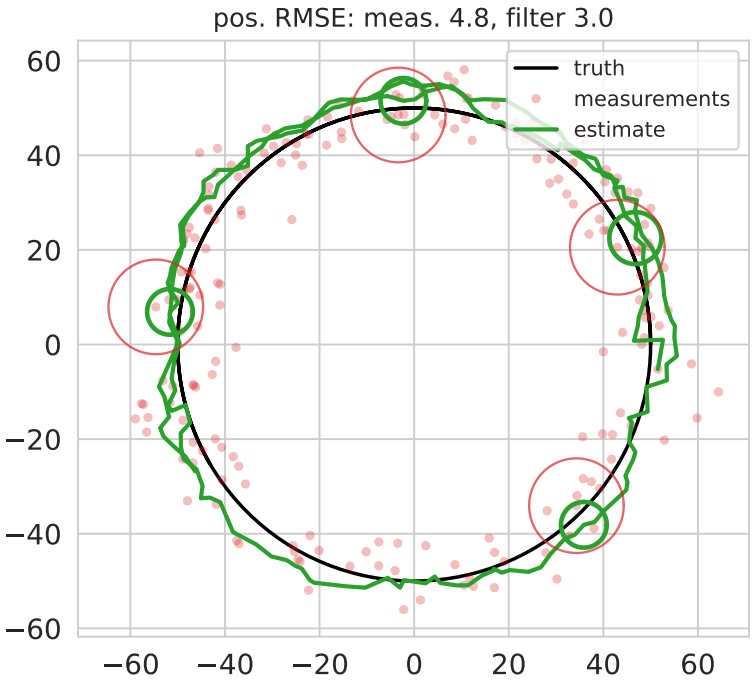
Observation: Only position is measured

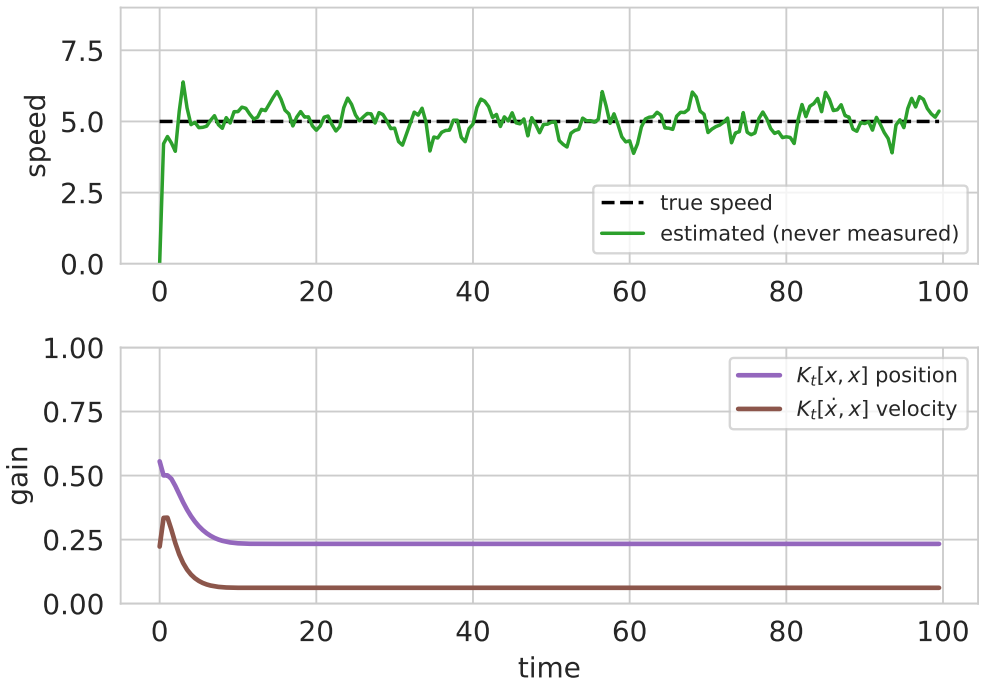
$$\mathbf{H}_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{R}_t = \begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix} \tag{14}$$

(x_t, y_t here are position coordinates, not the measurement vector $\mathbf{y}_t$.)

Example 2: Results

Circular trajectory, $\Delta t = 0.5$, $q = 0.5$, $\sigma_x = \sigma_y = 5$; the filter starts with zero velocity.





- Position error drops from 4.8 (measurements) to 3.0; filtered ellipses (green) are much smaller than a single measurement (red).
- **Velocity is estimated despite never being measured** (through Σ_{pv}).
- Gains converge quickly to steady state; the velocity gain is small but non-zero.

Computational Complexity

Per time step:

- **Prediction:**
  - Mean: $O(n_z^2)$ (matrix-vector multiply)
  - Covariance: $O(n_z^3)$ (matrix-matrix multiply)
- **Update:**
  - Innovation covariance $\mathbf{S}_t$: $O(n_z^2 n_y)$
  - Matrix inverse: $O(n_y^3)$ or $O(n_z^3)$ depending on form
  - Kalman gain: $O(n_z^2 n_y)$
  - Mean and covariance update: $O(n_z^2 n_y)$

Total: $O(n_z^3)$ per time step

Memory

Constant: $O(n_z^2)$ — only need to store current $\boldsymbol{\mu}_{t|t}$ and $\boldsymbol{\Sigma}_{t|t}$

Compare to batch inference: $O(T^3 n_z^3)$ complexity!

Key Properties of Kalman Filter

1. **Optimality:**
   - Minimum Mean Squared Error (MMSE) estimator
   - Best Linear Unbiased Estimator (BLUE)
   - Maximum A Posteriori (MAP) estimator for Gaussians
2. **Consistency:**
   - Innovation $\tilde{\mathbf{y}}_t$ should be white noise: $\tilde{\mathbf{y}}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_t)$
   - Covariance $\boldsymbol{\Sigma}_{t|t}$ represents true uncertainty (if model correct)
3. **Separability:**
   - Estimation and control can be designed independently
   - Basis for Linear Quadratic Gaussian (LQG) control

Limitations and Extensions

Kalman Filter Assumptions:

- Linear dynamics and observations
- Gaussian noise
- Known model parameters $(\mathbf{F}_t, \mathbf{H}_t, \mathbf{Q}_t, \mathbf{R}_t)$

What if assumptions are violated?

1. **Nonlinear systems:**
   - Extended Kalman Filter (EKF): linearize via Jacobians
   - Unscented Kalman Filter (UKF): sigma points
   - Particle Filter: Monte Carlo sampling
2. **Non-Gaussian noise:**
   - Mixture Kalman filters
   - Particle filters
3. **Unknown parameters:**
   - Expectation-Maximization (EM) algorithm
   - Adaptive Kalman filtering

Extended Kalman Filter (EKF)

Nonlinear system:

$$\mathbf{z}_t = f(\mathbf{z}_{t-1}, \mathbf{u}_t) + \mathbf{q}_t \tag{15}$$

$$\mathbf{y}_t = h(\mathbf{z}_t) + \mathbf{r}_t \tag{16}$$

Idea: Linearize around current estimate

Jacobians:

$$\mathbf{F}_t = \left. \frac{\partial f}{\partial \mathbf{z}} \right|_{\mathbf{z}=\boldsymbol{\mu}_{t-1|t-1}} \tag{17}$$

$$\mathbf{H}_t = \left. \frac{\partial h}{\partial \mathbf{z}} \right|_{\mathbf{z}=\boldsymbol{\mu}_{t|t-1}} \tag{18}$$

EKF equations: Same as Kalman filter, but use:

- $\boldsymbol{\mu}_{t|t-1} = f(\boldsymbol{\mu}_{t-1|t-1}, \mathbf{u}_t)$ instead of $\mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1}$
- $\hat{\mathbf{y}}_t = h(\boldsymbol{\mu}_{t|t-1})$ instead of $\mathbf{H}_t \boldsymbol{\mu}_{t|t-1}$
- $\mathbf{F}_t, \mathbf{H}_t$ computed via Jacobians

Works well for mildly nonlinear systems; can diverge if highly nonlinear.

Deep Kalman Filters

Idea: Combine Kalman filtering with deep learning

Approach:

- Parameterize $f(\cdot), h(\cdot)$ with neural networks
- Learn from data using gradient descent
- Use Kalman filter for inference given learned model

Architectures:

1. **Variational RNN (VRNN):**
   - $p(\mathbf{z}_t \mid \mathbf{z}_{t-1})$ and $p(\mathbf{y}_t \mid \mathbf{z}_t)$ are neural networks
   - Approximate inference with amortized variational inference
2. **Neural ODE + Kalman:**
   - Continuous-time dynamics: $\frac{d\mathbf{z}}{dt} = f_\theta(\mathbf{z}, t)$
   - Use Kalman filter for discrete observations

Active research area: combining model-based and learning-based approaches

Modern multi-object visual trackers

Tracker	Year	State estimator / motion model	Tracking highlights
SORT	2016	Linear KF (CV)	Fast baseline; short-term linear motion
DeepSORT	2017	Linear KF (CV)	Adds deep ReID; longer occlusion tolerance
Tracktor++	2019	Detector bbox regression	Uses detector regression as motion model; +CMC/ReID
CenterTrack	2020	Learned offsets (no KF)	Tracks centers via regressed inter-frame offsets
ByteTrack	2021	Linear KF (CV)	Improves continuity by using low-score detections
BoT-SORT	2022	KF + camera motion comp.	Improved box state + CMC for stable motion
StrongSORT	2022	KF + motion/camera corr.	Upgraded DeepSORT baseline; optional smoothing/link
OC-SORT	2023	KF + ORU re-update	Observation-centric re-update reduces KF drift in occlusion
TrackFormer	2022	Track queries (Transformer)	End-to-end tracking-by-attention; no explicit KF

Summary: Why Kalman Filter Matters

Theoretical Elegance

- Optimal Bayesian solution for linear-Gaussian systems
- Closed-form recursive equations
- Beautiful mathematical structure

Practical Impact

- Real-time state estimation with constant memory
- Navigation: GPS, inertial guidance, autonomous vehicles
- Control: LQG controllers, Model Predictive Control
- Signal processing: noise reduction, prediction

Conceptual Foundation

- Template for all filtering algorithms (EKF, UKF, particle filters)
- Connects to modern ML: RNNs, VAEs, S4/Mamba models
- Embodiment of Bayesian recursive inference

Key Takeaways

1. **Bayesian filter $\rightarrow$ Kalman filter:**
   - Predict (propagate uncertainty) $\rightarrow$ update (Bayes' rule)
   - Linear-Gaussian models keep everything Gaussian: track only $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$
2. **Covariance:**
   - Prediction reshapes ($\mathbf{F}_t$) and inflates ($\mathbf{Q}_t$); update shrinks
   - Correlations let us estimate states that are never measured
3. **Kalman Gain:**
   - Minimum-variance blend; depends only on relative uncertainty
   - Independent of the data; converges to a steady state
4. **Design:**
   - Large $\boldsymbol{\Sigma}_0$ is safe; $\boldsymbol{\mu}_0$ is forgotten
   - $\mathbf{Q}_t$, $\mathbf{R}_t$ set the bandwidth; check innovations for consistency
5. **Extensions:** EKF/UKF for nonlinear systems, RTS smoother for offline problems, deep learning connections

Further Reading

Classic References:

- R. E. Kalman (1960), "A New Approach to Linear Filtering and Prediction Problems," *ASME Journal of Basic Engineering*. (doi:10.1115/1.3662552)
- H. E. Rauch, F. Tung, C. T. Striebel (1965), "Maximum Likelihood Estimates of Linear Dynamic Systems," *AIAA Journal*. (RTS smoother; doi:10.2514/3.3166)

Textbooks:

- K. P. Murphy, *Probabilistic Machine Learning: Advanced Topics* (MIT Press, 2023), Ch. 8 (Gaussian filtering & smoothing).
- S. Särkkä & L. Svensson, *Bayesian Filtering and Smoothing*, 2nd ed. (Cambridge Univ. Press, 2023).
- D. Simon, *Optimal State Estimation: Kalman, H_∞, and Nonlinear Approaches* (Wiley, 2006).

Tutorials:

- G. Welch & G. Bishop, "An Introduction to the Kalman Filter," UNC TR 95-041 (1995; updated 2006).
- Y. Pei, S. Biswas, D. S. Fussell, K. Pingali, "An Elementary Introduction to Kalman Filtering," arXiv:1710.04055 (2017). (Also appears as an ACM tutorial article, 2019.)

Modern Extensions (SSM-based sequence models):

- A. Gu, K. Goel, C. Ré, "Efficiently Modeling Long Sequences with Structured State Spaces (S4)," ICLR 2022 (arXiv:2111.00396).
- A. Gu & T. Dao, "Mamba: Linear-Time Sequence Modeling with Selective State Spaces," arXiv:2312.00752 (2023).

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Appendix A

Derivation of the Bayesian filter (prediction and update)

A.1 Bayesian Filter Derivation: Model Assumptions

Goal: Compute $p(\mathbf{z}_t \mid \mathbf{y}_{1:t})$ recursively.

Starting point: joint probability factorization

For a state-space model (SSM), the joint distribution of states and observations up to time t factorizes as:

$$p(\mathbf{z}_{0:t}, \mathbf{y}_{1:t}) = p(\mathbf{z}_0) \prod_{k=1}^{t} p(\mathbf{z}_k \mid \mathbf{z}_{k-1})\, p(\mathbf{y}_k \mid \mathbf{z}_k).$$

Key assumptions

- **Markov property (state dynamics):**

$$p(\mathbf{z}_k \mid \mathbf{z}_{0:k-1}) = p(\mathbf{z}_k \mid \mathbf{z}_{k-1}).$$

- **Conditional independence (observations):**

$$p(\mathbf{y}_k \mid \mathbf{z}_{0:k}, \mathbf{y}_{1:k-1}) = p(\mathbf{y}_k \mid \mathbf{z}_k).$$

A.2 Bayesian Filter Derivation: Prediction

Prediction step (Chapman–Kolmogorov)

Start from the posterior at time $t-1$: $p(\mathbf{z}_{t-1} \mid \mathbf{y}_{1:t-1})$. To obtain the *prior* (predictive) at time t:

$$
\begin{aligned}
p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1}) &= \int p(\mathbf{z}_t, \mathbf{z}_{t-1} \mid \mathbf{y}_{1:t-1})\, d\mathbf{z}_{t-1} \\
&= \int p(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{y}_{1:t-1})\, p(\mathbf{z}_{t-1} \mid \mathbf{y}_{1:t-1})\, d\mathbf{z}_{t-1} \\
&\overset{\text{Markov}}{=} \int p(\mathbf{z}_t \mid \mathbf{z}_{t-1})\, p(\mathbf{z}_{t-1} \mid \mathbf{y}_{1:t-1})\, d\mathbf{z}_{t-1}.
\end{aligned}
$$

A.3 Bayesian Filter Derivation: Update

Update step (Bayes' rule)

Start with Bayes' theorem (posterior at time t): $p(\mathbf{z}_t \mid \mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_{1:t}|\mathbf{z}_t)\, p(\mathbf{z}_t)}{p(\mathbf{y}_{1:t})}$.

Decompose $\mathbf{y}_{1:t} = \{\mathbf{y}_{1:t-1}, \mathbf{y}_t\}$: $p(\mathbf{z}_t \mid \mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_t, \mathbf{y}_{1:t-1}|\mathbf{z}_t)\, p(\mathbf{z}_t)}{p(\mathbf{y}_t, \mathbf{y}_{1:t-1})}$.

Use conditional probability: $p(\mathbf{y}_t, \mathbf{y}_{1:t-1} \mid \mathbf{z}_t) = p(\mathbf{y}_t \mid \mathbf{y}_{1:t-1}, \mathbf{z}_t)\; p(\mathbf{y}_{1:t-1} \mid \mathbf{z}_t)$.

Conditional independence assumption: $p(\mathbf{y}_t \mid \mathbf{y}_{1:t-1}, \mathbf{z}_t) = p(\mathbf{y}_t \mid \mathbf{z}_t)$.

Substitute into Bayes: $p(\mathbf{z}_t \mid \mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_t|\mathbf{z}_t)\; p(\mathbf{y}_{1:t-1}|\mathbf{z}_t)\; p(\mathbf{z}_t)}{p(\mathbf{y}_t, \mathbf{y}_{1:t-1})}$.

Recognize two identities: $\frac{p(\mathbf{y}_{1:t-1}|\mathbf{z}_t)\, p(\mathbf{z}_t)}{p(\mathbf{y}_{1:t-1})} = p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1})$, $\qquad \frac{p(\mathbf{y}_{1:t-1})}{p(\mathbf{y}_t, \mathbf{y}_{1:t-1})} = \frac{1}{p(\mathbf{y}_t|\mathbf{y}_{1:t-1})}$.

Therefore (Bayes update): $p(\mathbf{z}_t \mid \mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_t|\mathbf{z}_t)\; p(\mathbf{z}_t|\mathbf{y}_{1:t-1})}{p(\mathbf{y}_t|\mathbf{y}_{1:t-1})}$.

Appendix B

Kalman filter equations from the Gaussian identities in tabular form (same content as Bridges 1–3), optimality of the gain, steady-state gain, and the information form (Corollary 1)

B.1 Kalman Filter: Prediction (Time Update)

Goal: derive $p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1})$.

Mapping: Gaussian formulas

Use Corollary 2: $p(\mathbf{y}) = \int p(\mathbf{y} \mid \mathbf{x}) p(\mathbf{x})\, d\mathbf{x}$.

Theorem symbol	Kalman symbol
$\mathbf{x}$	$\mathbf{z}_{t-1}$
$\mathbf{y}$	$\mathbf{z}_t$
A	$\mathbf{F}_t$
b	$\mathbf{B}_t\mathbf{u}_t + \mathbf{b}_t$
$\Sigma_{\mathbf{y}\mid\mathbf{x}}$	$\mathbf{Q}_t$
$\mu_{\mathbf{x}}$	$\boldsymbol{\mu}_{t-1\mid t-1}$
$\Sigma_{\mathbf{x}}$	$\boldsymbol{\Sigma}_{t-1\mid t-1}$

Intuition: Affine transformation shifts the mean by $\mathbf{F}_t(\cdot) + (\mathbf{B}_t\mathbf{u}_t + \mathbf{b}_t)$ and adds uncertainty: *propagated covariance* + $\mathbf{Q}_t$.

Theorem 3 + Corollary 2

If $p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu_{\mathbf{x}}, \Sigma_{\mathbf{x}})$ and

$p(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(\mathbf{y}; A\mathbf{x} + b, \Sigma_{\mathbf{y}\mid\mathbf{x}})$,

then the marginal (predictive) distribution is

$$p(\mathbf{y}) = \mathcal{N}\big(\mathbf{y};\ A\mu_{\mathbf{x}} + b,\ \Sigma_{\mathbf{y}\mid\mathbf{x}} + A\Sigma_{\mathbf{x}}A^{\top}\big).$$

Apply mapping $\Rightarrow$ Kalman prediction

$$p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1}) = \mathcal{N}\big(\mathbf{z}_t; \boldsymbol{\mu}_{t\mid t-1}, \boldsymbol{\Sigma}_{t\mid t-1}\big)$$

$$\boldsymbol{\mu}_{t\mid t-1} = \mathbf{F}_t\boldsymbol{\mu}_{t-1\mid t-1} + \mathbf{B}_t\mathbf{u}_t + \mathbf{b}_t$$

$$\boldsymbol{\Sigma}_{t\mid t-1} = \mathbf{Q}_t + \mathbf{F}_t\boldsymbol{\Sigma}_{t-1\mid t-1}\mathbf{F}_t^{\top}$$

B.2 Kalman Filter: Update (1/2) – Build a Joint Gaussian

Given (from prediction): $p(\mathbf{z}_t \mid \mathbf{y}_{1:t-1}) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1})$.

Mapping to Theorem 3

Theorem symbol	Kalman symbol
$\mathbf{x}$	$\mathbf{z}_t$
$\mathbf{y}$	$\mathbf{y}_t$
A	$\mathbf{H}_t$
b	$\mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t$
$\Sigma_{\mathbf{y}\|\mathbf{x}}$	$\mathbf{R}_t$
$\mu_{\mathbf{x}}$	$\boldsymbol{\mu}_{t\|t-1}$
$\Sigma_{\mathbf{x}}$	$\boldsymbol{\Sigma}_{t\|t-1}$

Observation model (Affine-Gaussian)

$$p(\mathbf{y}_t \mid \mathbf{z}_t) = \mathcal{N}(\mathbf{y}_t; \mathbf{H}_t\mathbf{z}_t + \mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t, \mathbf{R}_t).$$

Why we do this Once we have the **joint Gaussian** of $(\mathbf{z}_t, \mathbf{y}_t)$, Theorem 2 gives the **conditional** $p(\mathbf{z}_t \mid \mathbf{y}_t, \mathbf{y}_{1:t-1})$.

Theorem 3 (Affine-Gaussian)

If $p(\mathbf{x}) = \mathcal{N}(\mu_{\mathbf{x}}, \Sigma_{\mathbf{x}})$ and $p(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(A\mathbf{x} + b, \Sigma_{\mathbf{y}|\mathbf{x}})$, then $\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}$ is jointly Gaussian with block covariance structure.

Apply mapping $\Rightarrow$ joint of $(\mathbf{z}_t, \mathbf{y}_t) \mid \mathbf{y}_{1:t-1}$

$$\begin{bmatrix}\mathbf{z}_t\\ \mathbf{y}_t\end{bmatrix} \Big| \mathbf{y}_{1:t-1} \sim \mathcal{N}\left(\begin{bmatrix}\boldsymbol{\mu}_{t|t-1}\\ \hat{\mathbf{y}}_t\end{bmatrix}, \begin{bmatrix}\boldsymbol{\Sigma}_{t|t-1} & \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top\\ \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1} & \mathbf{S}_t\end{bmatrix}\right)$$

$$\hat{\mathbf{y}}_t = \mathbb{E}[\mathbf{y}_t \mid \mathbf{y}_{1:t-1}] = \mathbf{H}_t\boldsymbol{\mu}_{t|t-1} + \mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t$$

$$\mathbf{S}_t = \mathrm{Cov}(\mathbf{y}_t \mid \mathbf{y}_{1:t-1}) = \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top + \mathbf{R}_t$$

B.3 Kalman Filter: Update (2/2) – Condition the Joint

Goal: Compute the filtering posterior: $p(\mathbf{z}_t \mid \mathbf{y}_{1:t}) = p(\mathbf{z}_t \mid \mathbf{y}_t, \mathbf{y}_{1:t-1})$.

Theorem 2

For a joint Gaussian $\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}$,

$$\boldsymbol{\mu}_{\mathbf{x}|\mathbf{y}} = \boldsymbol{\mu}_{\mathbf{x}} + \boldsymbol{\Sigma}_{\mathbf{xy}}\boldsymbol{\Sigma}_{\mathbf{yy}}^{-1}(\mathbf{y} - \boldsymbol{\mu}_{\mathbf{y}}),$$

$$\boldsymbol{\Sigma}_{\mathbf{x}|\mathbf{y}} = \boldsymbol{\Sigma}_{\mathbf{xx}} - \boldsymbol{\Sigma}_{\mathbf{xy}}\boldsymbol{\Sigma}_{\mathbf{yy}}^{-1}\boldsymbol{\Sigma}_{\mathbf{yx}}.$$

Mapping to joint from Update

Theorem 2 symbol	Kalman symbol
$\mathbf{x}$	$\mathbf{z}_t$
$\mathbf{y}$	$\mathbf{y}_t$
$\boldsymbol{\mu}_{\mathbf{x}}$	$\boldsymbol{\mu}_{t\|t-1}$
$\boldsymbol{\mu}_{\mathbf{y}}$	$\hat{\mathbf{y}}_t$
$\boldsymbol{\Sigma}_{\mathbf{xy}}$	$\boldsymbol{\Sigma}_{t\|t-1}\mathbf{H}_t^\top$
$\boldsymbol{\Sigma}_{\mathbf{yy}}$	$\mathbf{S}_t$

Kalman update = Theorem 2

$$\mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top \mathbf{S}_t^{-1}$$

$$\mathbf{S}_t = \mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top + \mathbf{R}_t$$

Define the **innovation** (prediction error):

$$\tilde{\mathbf{y}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$$

$$(\hat{\mathbf{y}}_t = \mathbf{H}_t\boldsymbol{\mu}_{t|t-1} + \mathbf{D}_t\mathbf{u}_t + \mathbf{d}_t)$$

$$\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t\,\tilde{\mathbf{y}}_t$$

$$\begin{aligned}\boldsymbol{\Sigma}_{t|t} &= \boldsymbol{\Sigma}_{t|t-1} - \mathbf{K}_t\,\mathbf{S}_t\,\mathbf{K}_t^\top \\ &= (\mathbf{I} - \mathbf{K}_t\mathbf{H}_t)\boldsymbol{\Sigma}_{t|t-1}\end{aligned}$$

B.4 Optimality of the Gain (Matrix Form)

Consider *any* linear update $\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}(\mathbf{y}_t - \hat{\mathbf{y}}_t)$. The estimation error is

$$\mathbf{e}_{t|t} = (\mathbf{I} - \mathbf{K}\mathbf{H}_t)\,\mathbf{e}_{t|t-1} - \mathbf{K}\,\mathbf{r}_t,$$

with $\mathbf{e}_{t|t-1}$ and $\mathbf{r}_t$ independent, so its covariance is (Joseph form)

$$\boldsymbol{\Sigma}_{t|t}(\mathbf{K}) = (\mathbf{I} - \mathbf{K}\mathbf{H}_t)\,\boldsymbol{\Sigma}_{t|t-1}(\mathbf{I} - \mathbf{K}\mathbf{H}_t)^\top + \mathbf{K}\mathbf{R}_t\mathbf{K}^\top.$$

Minimize the mean squared error $\operatorname{tr}\boldsymbol{\Sigma}_{t|t}(\mathbf{K})$:

$$\frac{\partial \operatorname{tr}\boldsymbol{\Sigma}_{t|t}}{\partial \mathbf{K}} = -2(\mathbf{I} - \mathbf{K}\mathbf{H}_t)\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top + 2\mathbf{K}\mathbf{R}_t = \mathbf{0}$$

$$\Rightarrow \quad \mathbf{K}\underbrace{(\mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top + \mathbf{R}_t)}_{\mathbf{S}_t} = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top$$

$$\Rightarrow \quad \mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}_t^\top\mathbf{S}_t^{-1}, \qquad \boldsymbol{\Sigma}_{t|t}(\mathbf{K}_t) = (\mathbf{I} - \mathbf{K}_t\mathbf{H}_t)\boldsymbol{\Sigma}_{t|t-1}.$$

- Scalar case ($H_t = 1$): $\sigma^2_{t|t}(K) = (1-K)^2\sigma^2_{t|t-1} + K^2 r_t$, as on the main slides.
- The Joseph form is valid for any $\mathbf{K}$ and is numerically more robust than $(\mathbf{I} - \mathbf{K}_t\mathbf{H}_t)\boldsymbol{\Sigma}_{t|t-1}$.

B.5 Steady-State Gain of the Scalar Random Walk

For $F_t = H_t = 1$, $q_t = q$, $r_t = r$, the predicted variance $P_t = \sigma^2_{t|t-1}$ obeys the (scalar) Riccati recursion

$$P_{t+1} = \underbrace{(1 - K_t)P_t}_{\sigma^2_{t|t}} + q = \frac{P_t\, r}{P_t + r} + q.$$

At the fixed point $P_{t+1} = P_t = P$:

$$P(P + r) = P\, r + q(P + r) \;\Rightarrow\; P^2 - qP - qr = 0$$

$$\Rightarrow \quad P = \frac{q + \sqrt{q^2 + 4qr}}{2}, \qquad K_\infty = \frac{P}{P + r}.$$

- Example: $q = 0.1,\ r = 1 \Rightarrow P = 0.370,\ K_\infty = 0.270$ (dashed line in the gain-evolution figure).
- $q = r \Rightarrow K_\infty = (\sqrt{5} - 1)/2 \approx 0.62;\ \ q \to 0 \Rightarrow K_\infty \to 0.$
- The map $P_t \mapsto P_{t+1}$ is increasing with slope $r^2/(P_t + r)^2 < 1$, so it converges to P from any σ_0^2.
- Matrix case: Discrete Algebraic Riccati Equation (DARE), solved numerically (e.g. `scipy.linalg.solve_discrete_are`).

B.6 Alternative Update via Corollary 1 (Information Form)

Corollary 1 with the renaming of Bridge 2 ($\boldsymbol{x}=\mathbf{z}_t$, $\boldsymbol{y}=\mathbf{y}_t$, $\boldsymbol{A}=\mathbf{H}_t$, $\boldsymbol{b}=\mathbf{D}_t\mathbf{u}_t+\mathbf{d}_t$, $\Sigma_{y|x}=\mathbf{R}_t$, $\boldsymbol{\mu}_x=\boldsymbol{\mu}_{t|t-1}$, $\Sigma_x=\boldsymbol{\Sigma}_{t|t-1}$):

Information form of the update

$$\boldsymbol{\Sigma}_{t|t}=\left(\boldsymbol{\Sigma}_{t|t-1}^{-1}+\mathbf{H}_t^\top\mathbf{R}_t^{-1}\mathbf{H}_t\right)^{-1},\qquad \boldsymbol{\mu}_{t|t}=\boldsymbol{\Sigma}_{t|t}\left(\boldsymbol{\Sigma}_{t|t-1}^{-1}\boldsymbol{\mu}_{t|t-1}+\mathbf{H}_t^\top\mathbf{R}_t^{-1}(\mathbf{y}_t-\mathbf{D}_t\mathbf{u}_t-\mathbf{d}_t)\right)$$

Same as the Kalman update:

- Covariance (Woodbury identity):
  $\left(\boldsymbol{\Sigma}^{-1}+\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}=\boldsymbol{\Sigma}-\boldsymbol{\Sigma}\mathbf{H}^\top(\mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^\top+\mathbf{R})^{-1}\mathbf{H}\boldsymbol{\Sigma}=(\mathbf{I}-\mathbf{K}_t\mathbf{H}_t)\boldsymbol{\Sigma}_{t|t-1}.$
- Gain identity:
  $\boldsymbol{\Sigma}_{t|t}\mathbf{H}_t^\top\mathbf{R}_t^{-1}=\boldsymbol{\Sigma}\mathbf{H}^\top\mathbf{R}^{-1}-\boldsymbol{\Sigma}\mathbf{H}^\top\mathbf{S}^{-1}\mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^\top\mathbf{R}^{-1}=\boldsymbol{\Sigma}\mathbf{H}^\top\mathbf{S}^{-1}(\mathbf{S}-\mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^\top)\mathbf{R}^{-1}=\mathbf{K}_t.$
- Mean: $\boldsymbol{\mu}_{t|t}=(\mathbf{I}-\mathbf{K}_t\mathbf{H}_t)\boldsymbol{\mu}_{t|t-1}+\mathbf{K}_t(\mathbf{y}_t-\mathbf{D}_t\mathbf{u}_t-\mathbf{d}_t)=\boldsymbol{\mu}_{t|t-1}+\mathbf{K}_t(\mathbf{y}_t-\hat{\mathbf{y}}_t).$

Interpretation: precisions add (prior $\boldsymbol{\Sigma}_{t|t-1}^{-1}$ + measurement $\mathbf{H}_t^\top\mathbf{R}_t^{-1}\mathbf{H}_t$): the *information filter*, convenient for many measurements or a diffuse prior. In 1D it is the Multiplication rule.